

GLOBAL FOOTBALL ANALYSIS

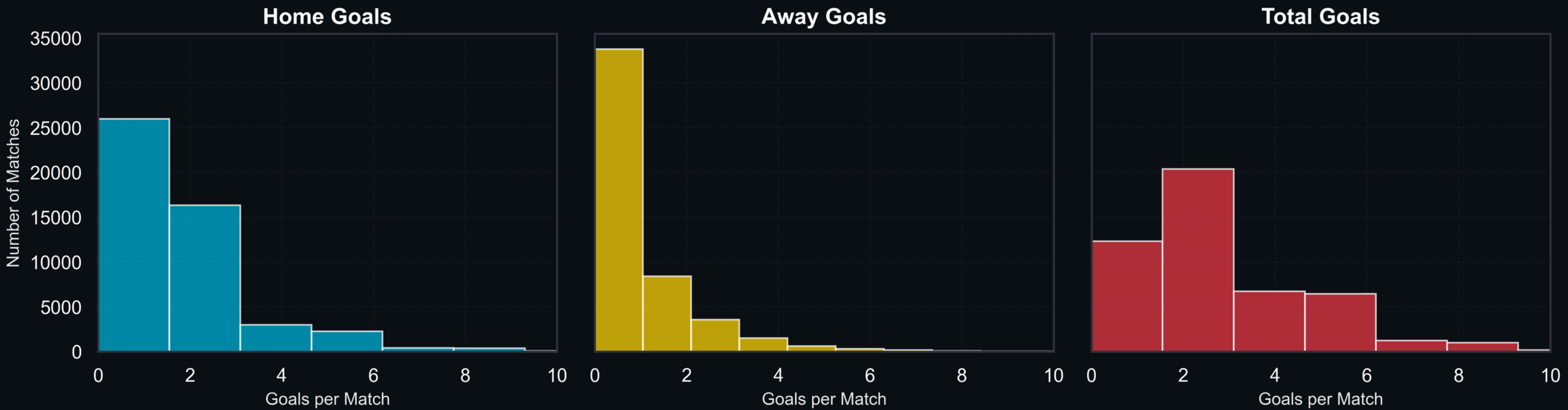
A Data-Driven Exploration of 150 Years of International Football

This project visualizes over 40,000 international matches from 1872 to 2023 — uncovering key patterns in scoring, competition balance, and the evolution of global football.

Author: Sezer Bayraktar | Tools: Python (Numpy, Pandas, Matplotlib, Seaborn)

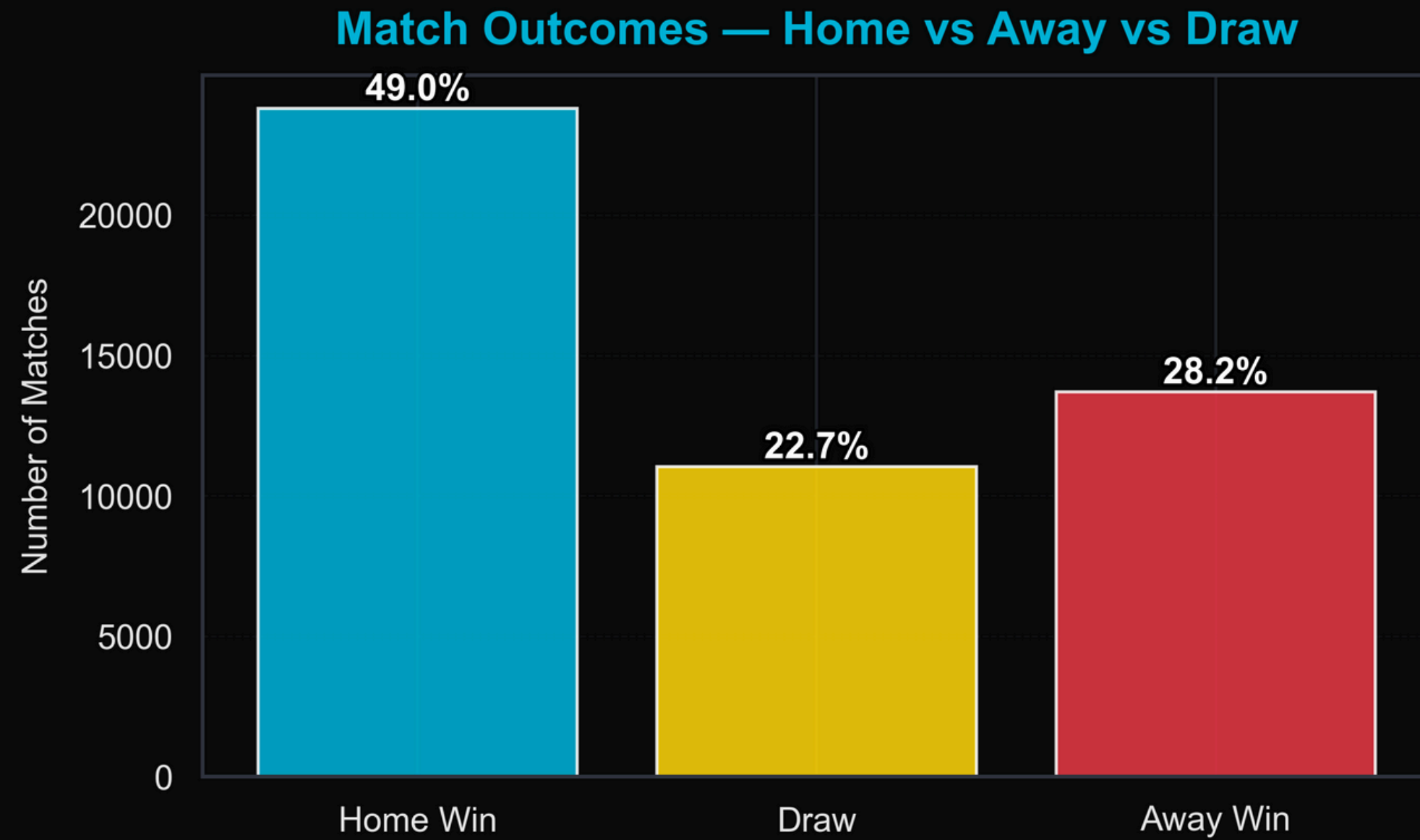
Dataset: kaggle.com/datasets/martj42/international-football-results-from-1872-to-2017

Goal Distribution — Home vs Away vs Total



Home teams tend to score more goals. Most matches end with 2–3 total goals — highlighting football's tactical and low-scoring nature.

This visualization compares the frequency of goals scored by home teams, away teams, and in total across all matches. The distributions reveal that home teams generally score more goals, and most matches end with only 2–3 total goals — emphasizing football’s inherently tactical and low-scoring nature.

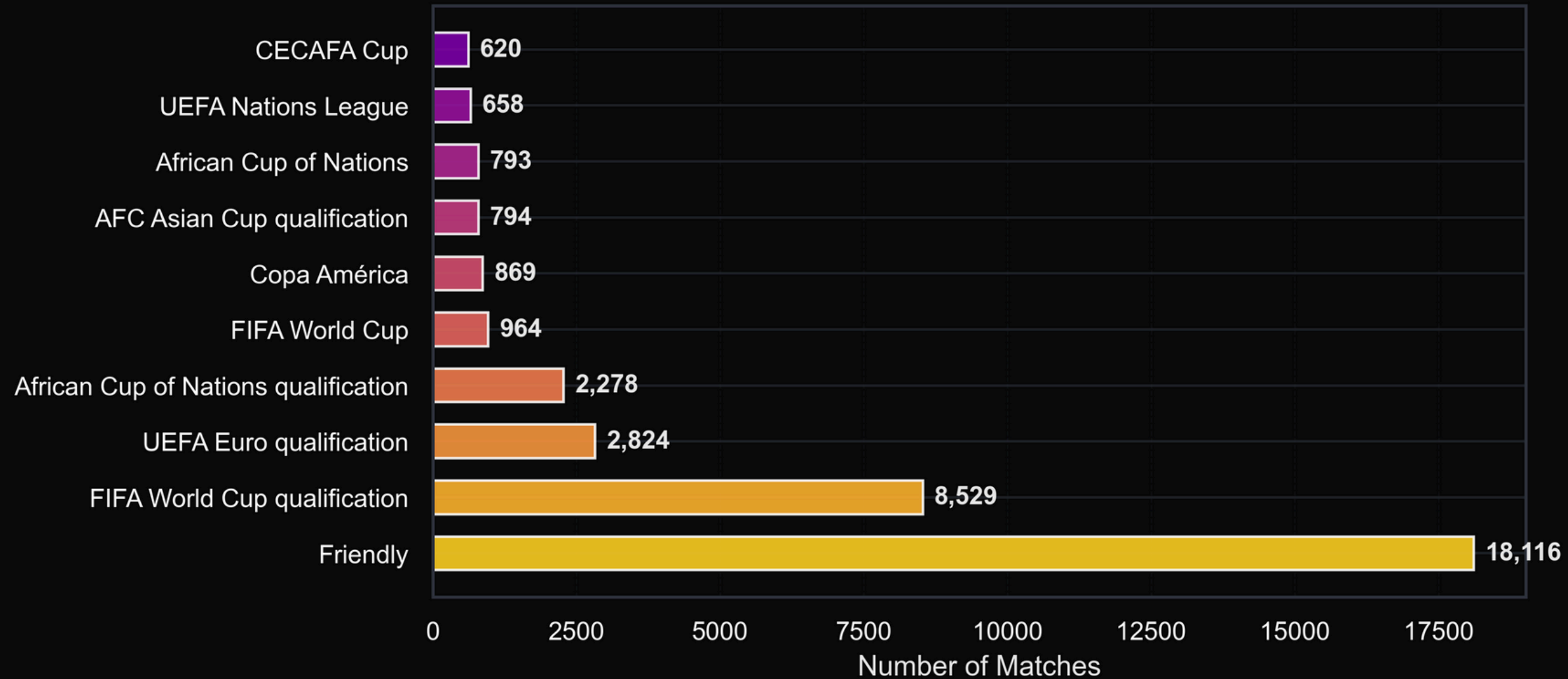


Home teams win nearly half of all matches — clear evidence of home-field advantage in international football.

This bar chart visualizes the overall match outcomes, revealing a strong home-field advantage in international football. Nearly half of all matches (49%) are won by home teams, while away wins account for 28% and draws for 23%. The data confirms that playing on home soil provides a significant competitive edge.

Top 10 Tournaments by Match Count

Friendly matches account for nearly half of all recorded international games — showing that most football history is built on non-tournament play.



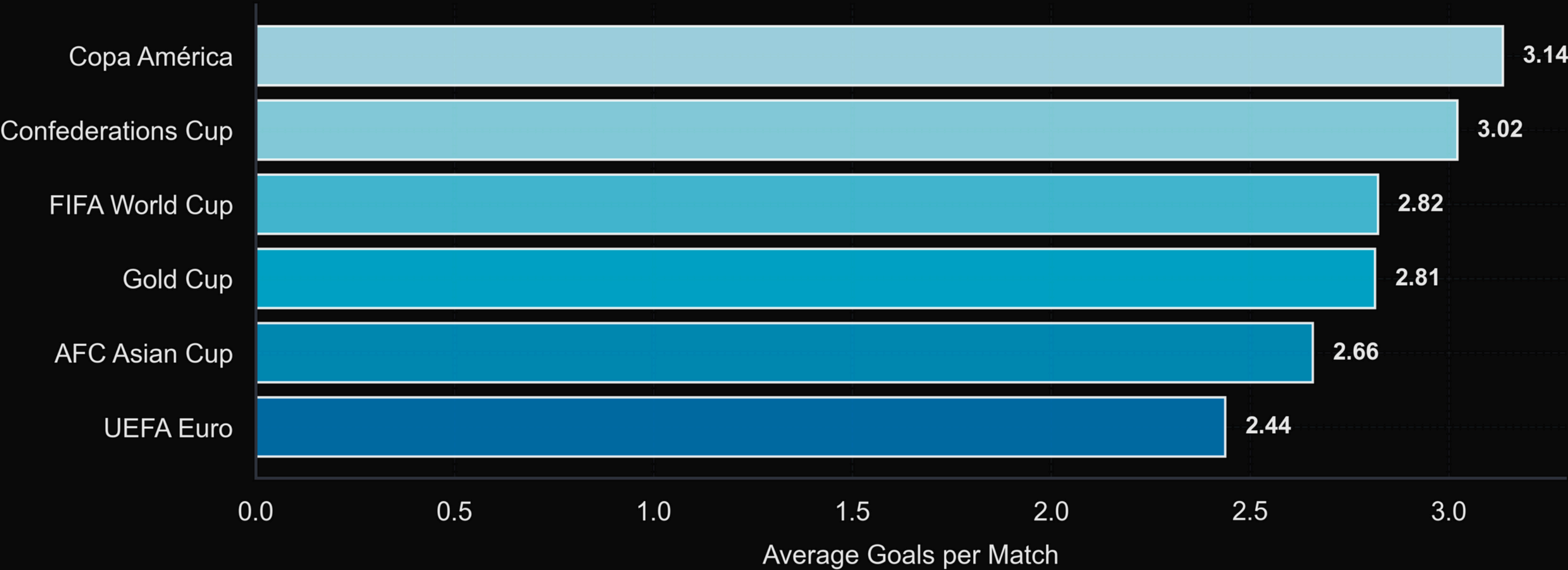
This horizontal bar chart highlights the tournaments with the highest number of recorded matches. Friendly games dominate the dataset with over 18,000 matches — nearly half of all international games — indicating that much of football’s global history has been shaped outside official tournaments. Qualifiers such as World Cup and Euro qualifications also account for a major share of international play.

Total International Matches Played by Year (1872–2020)



This time series visualizes the evolution of international football from its origins in the 19th century to the modern era. The sharp declines during WWI and WWII reflect the global disruption of competition, while the post-1950 boom illustrates football's transformation into a truly global sport. The 5-year rolling average smooths annual variations, revealing long-term growth patterns.

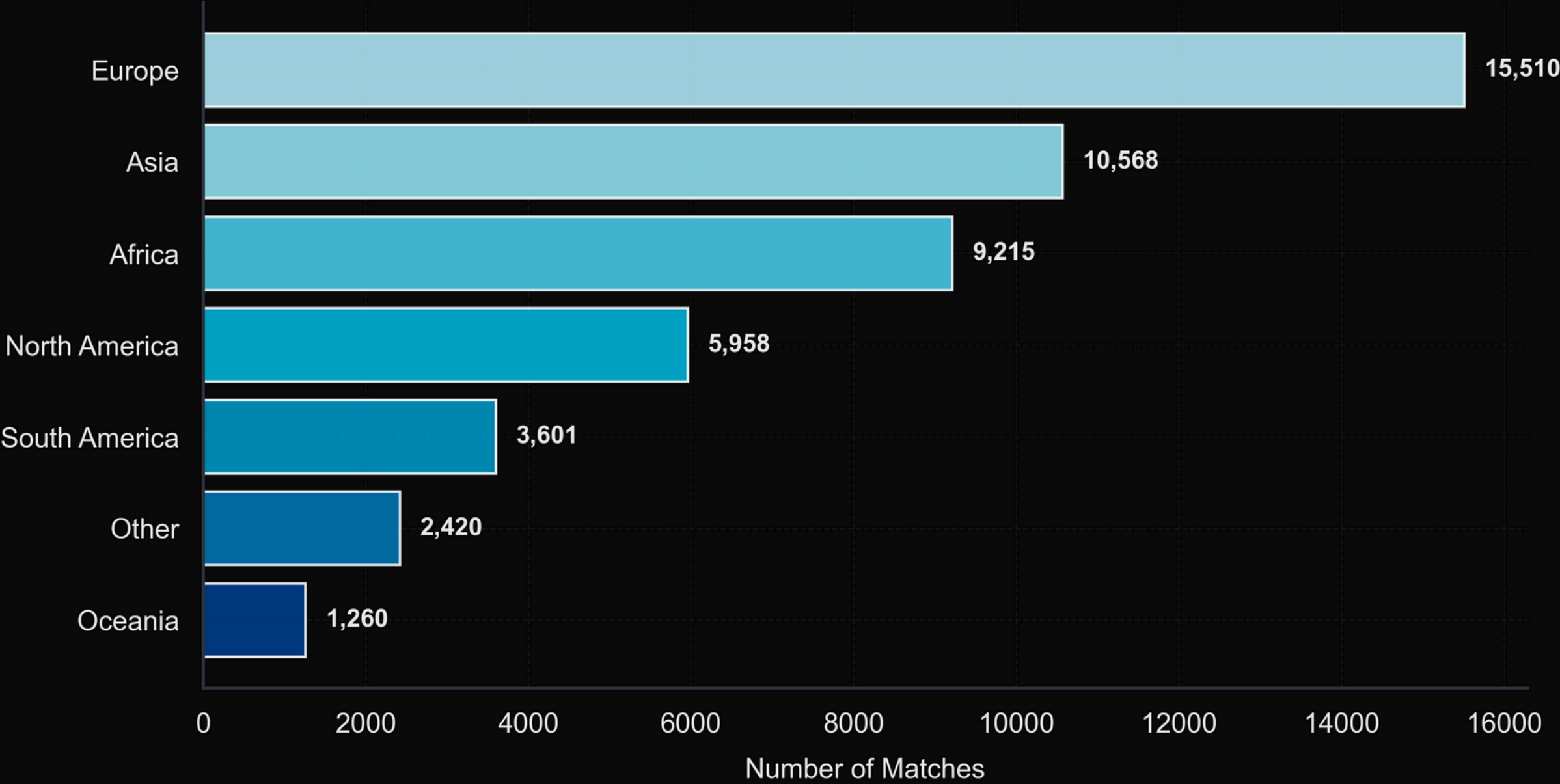
Goals per Match — Major International Tournaments



Only major international tournaments included — illustrating goal-scoring trends across elite competitions.

This chart compares the average number of goals scored per match across major international tournaments. Copa América leads with 3.14 goals per game, followed closely by the Confederations Cup and the World Cup. In contrast, the UEFA Euro tends to be more tactical and defensively balanced, averaging just 2.44 goals per match — illustrating stylistic diversity across global competitions.

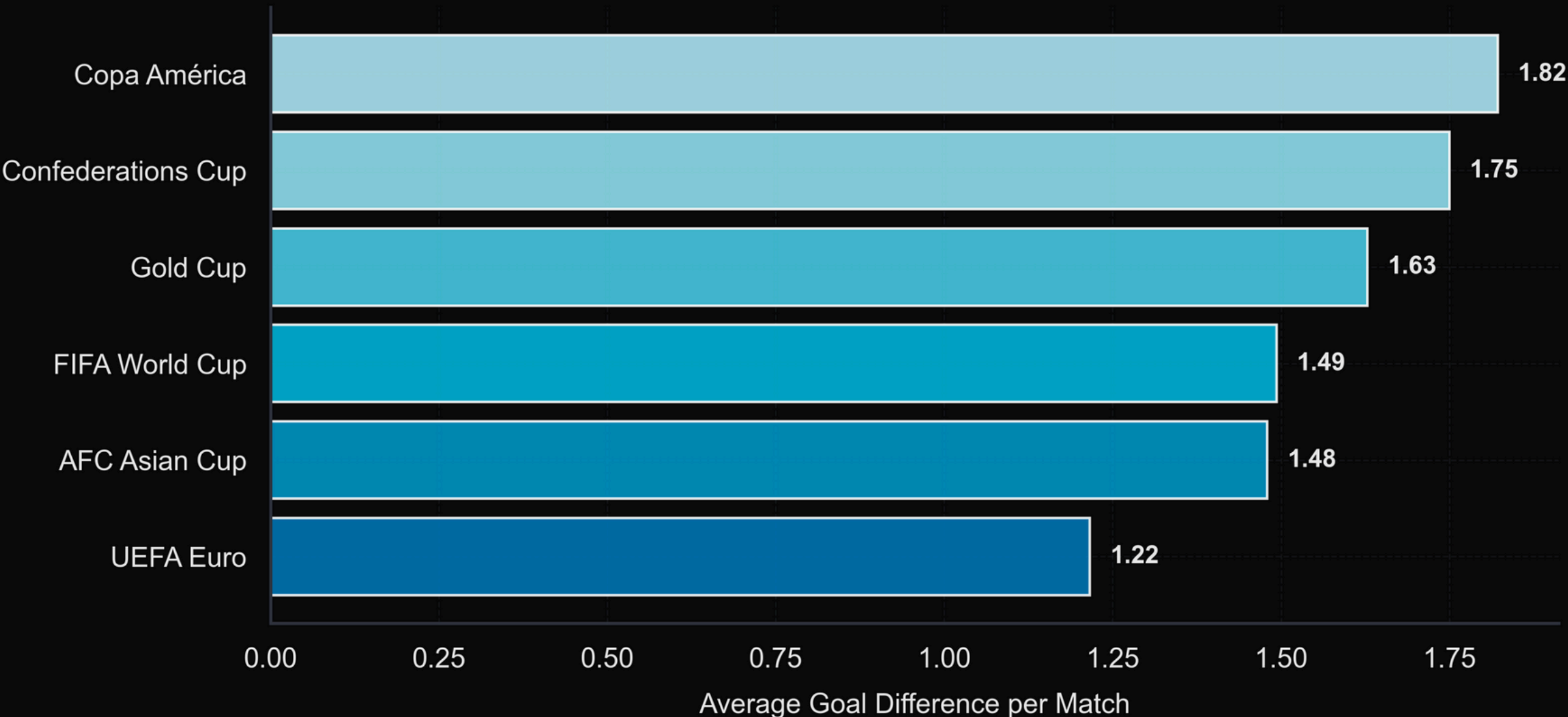
International Matches by Continent (Home Team Region)



Europe leads global football activity, followed by Asia and Africa — reflecting both historical dominance and rapid modern expansion.

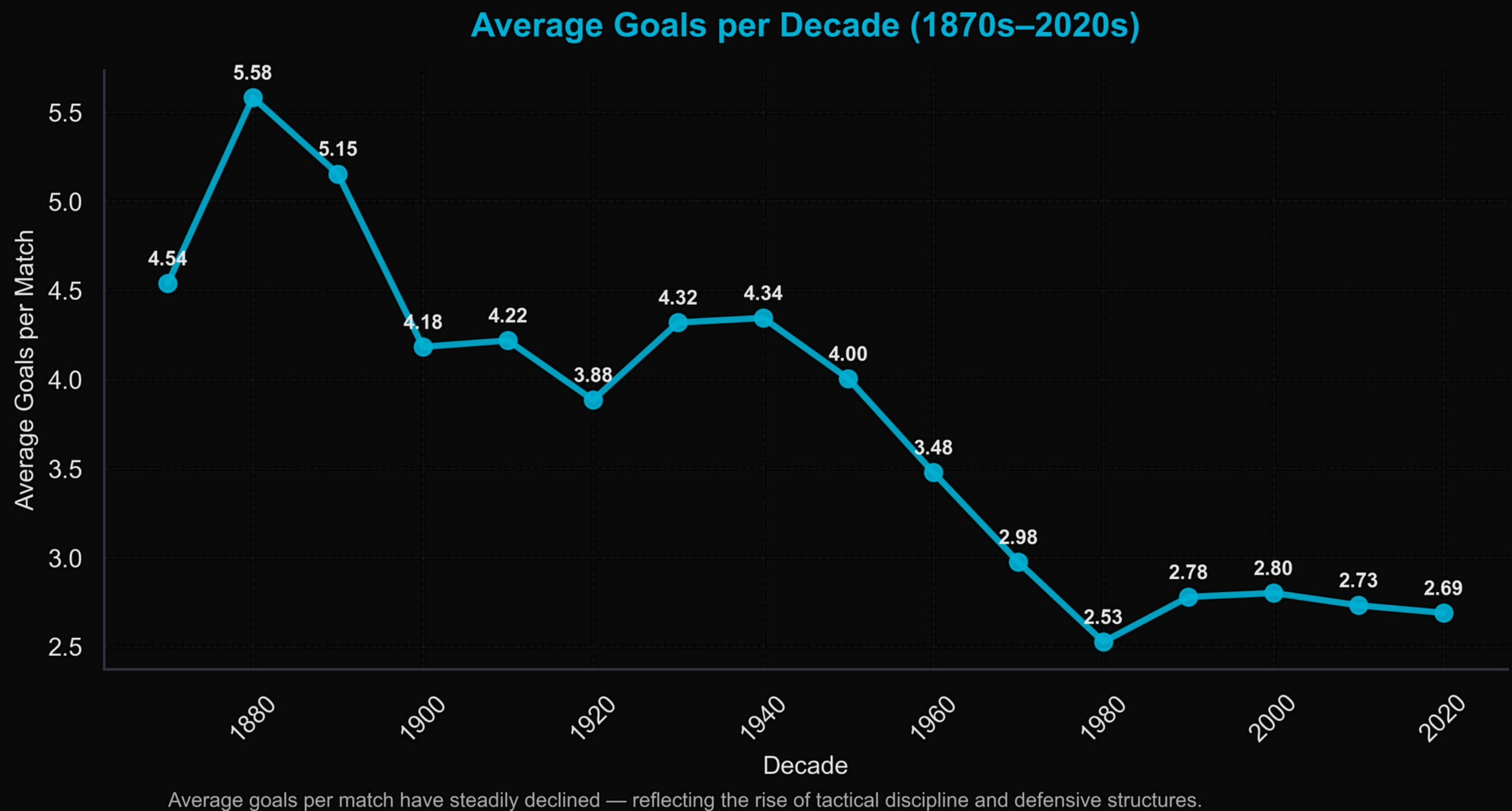
This chart illustrates the distribution of international matches by continent, based on the home team’s region. Europe dominates global football activity with over 15,000 matches — a result of both historical prominence and well-developed infrastructure. Asia and Africa follow, reflecting the sport’s rapid modern expansion across emerging football markets.

Average Goal Difference by Major Tournament

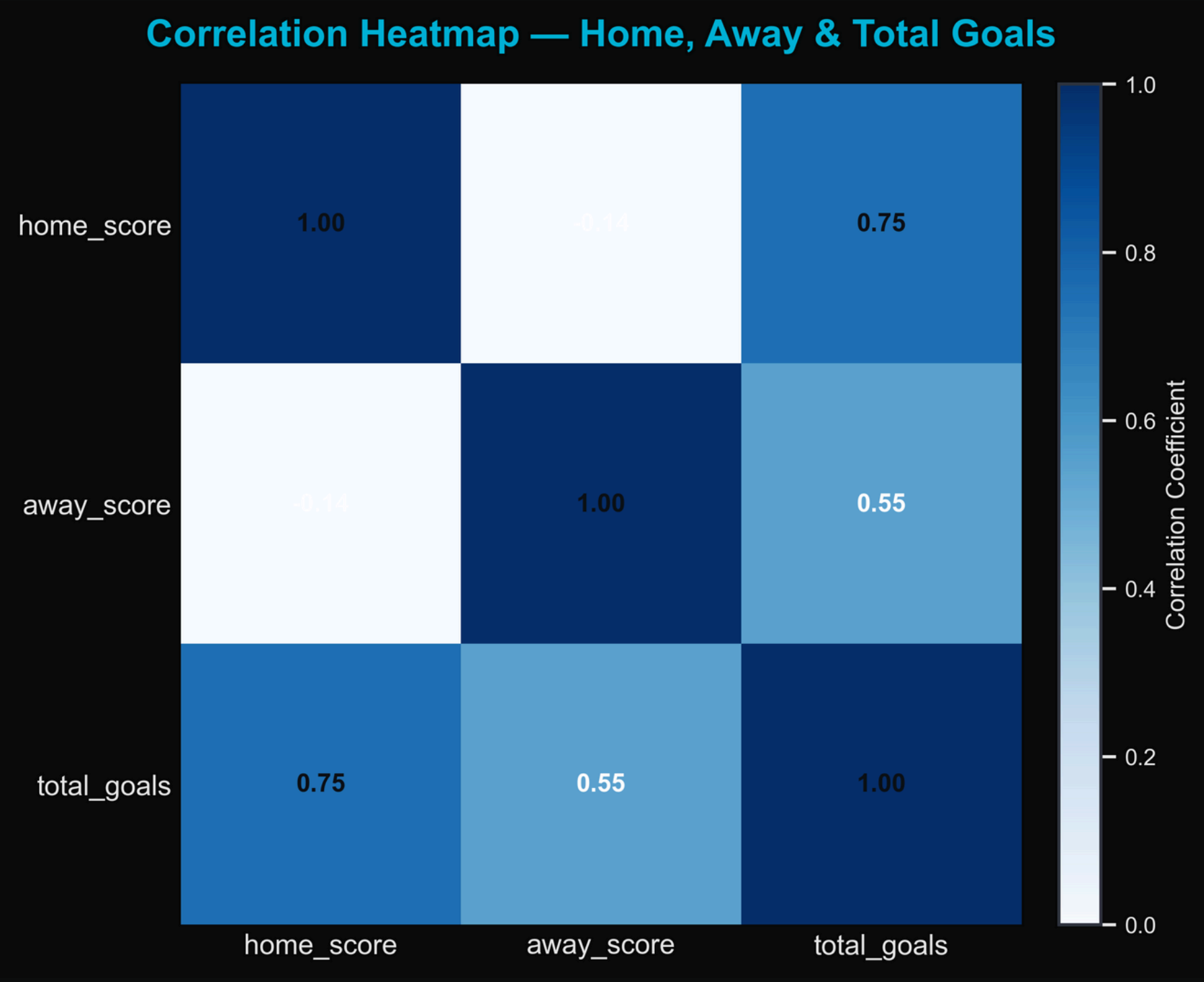


Smaller average goal differences indicate more balanced and competitive tournaments.

This visualization compares the average goal difference per match across major international tournaments. Smaller goal differences reflect tighter, more competitive matches. The UEFA Euro stands out as the most balanced tournament (1.22), while Copa América shows the widest margins (1.82), indicating greater disparity in team performance across South America.



This line chart tracks the evolution of goal-scoring across 150 years of international football. The average goals per match have steadily declined from over 5 in the late 19th century to under 3 in recent decades — illustrating the rise of tactical discipline, defensive organization, and the increasing balance between teams in modern football.



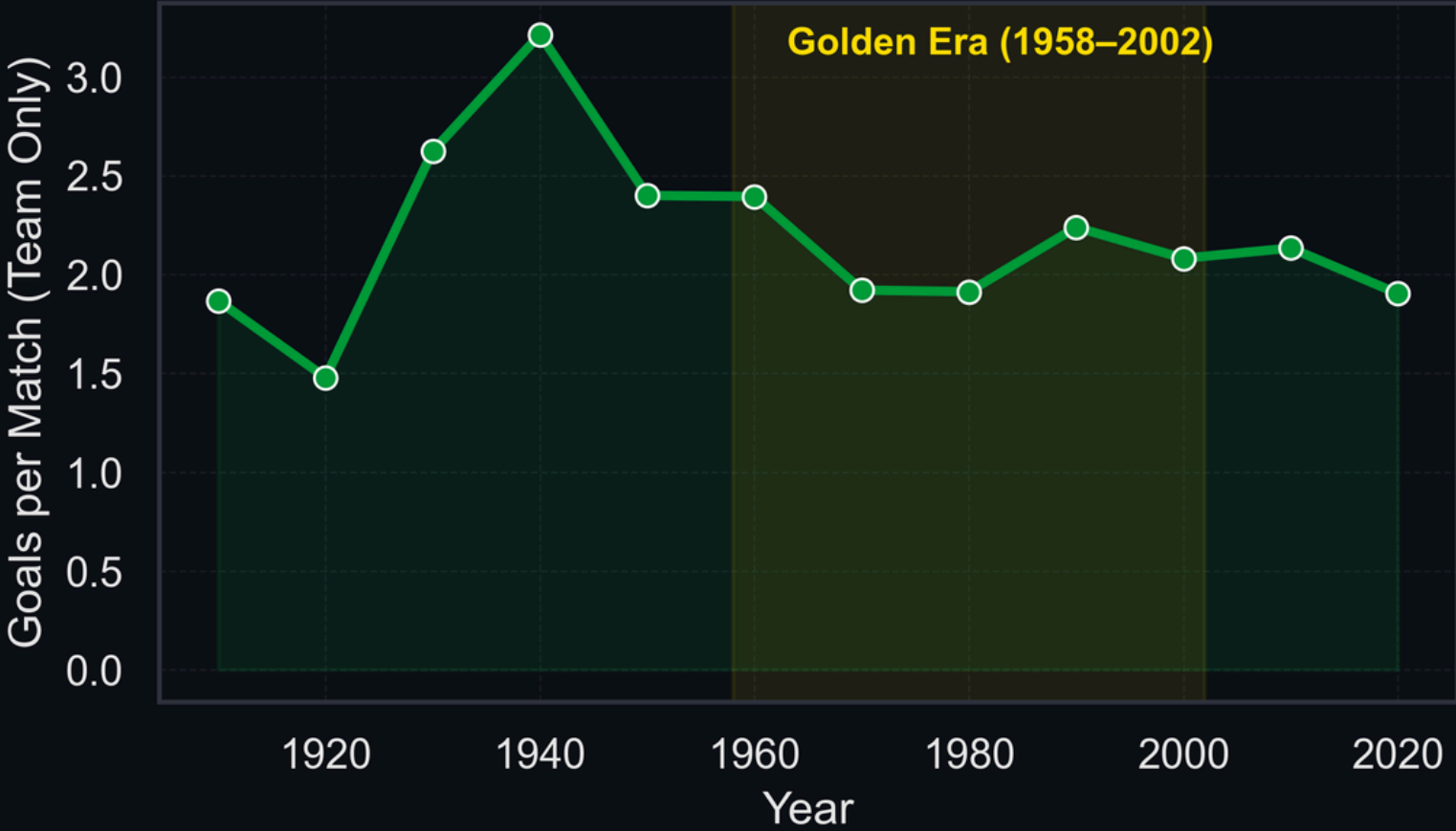
This heatmap visualizes the relationships between home goals, away goals, and total goals. A strong positive correlation exists between total goals and both home (0.75) and away (0.55) scores, confirming their combined influence on match outcomes. The weak negative correlation between home and away scores (−0.14) suggests that dominant home performances often limit away scoring.

From Global Trends to National Insights

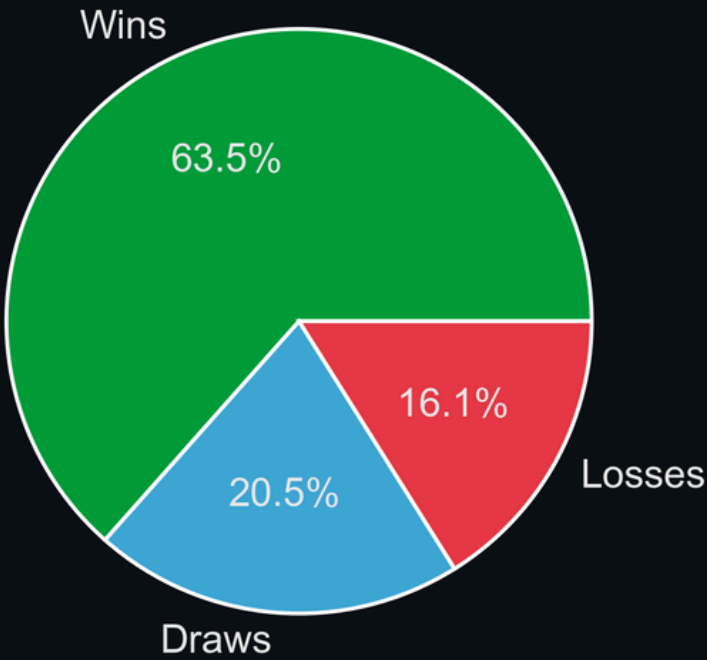
After exploring worldwide football dynamics, the analysis now shifts focus to individual nations — highlighting their performance, rivalries, and evolution across history.

BRAZIL

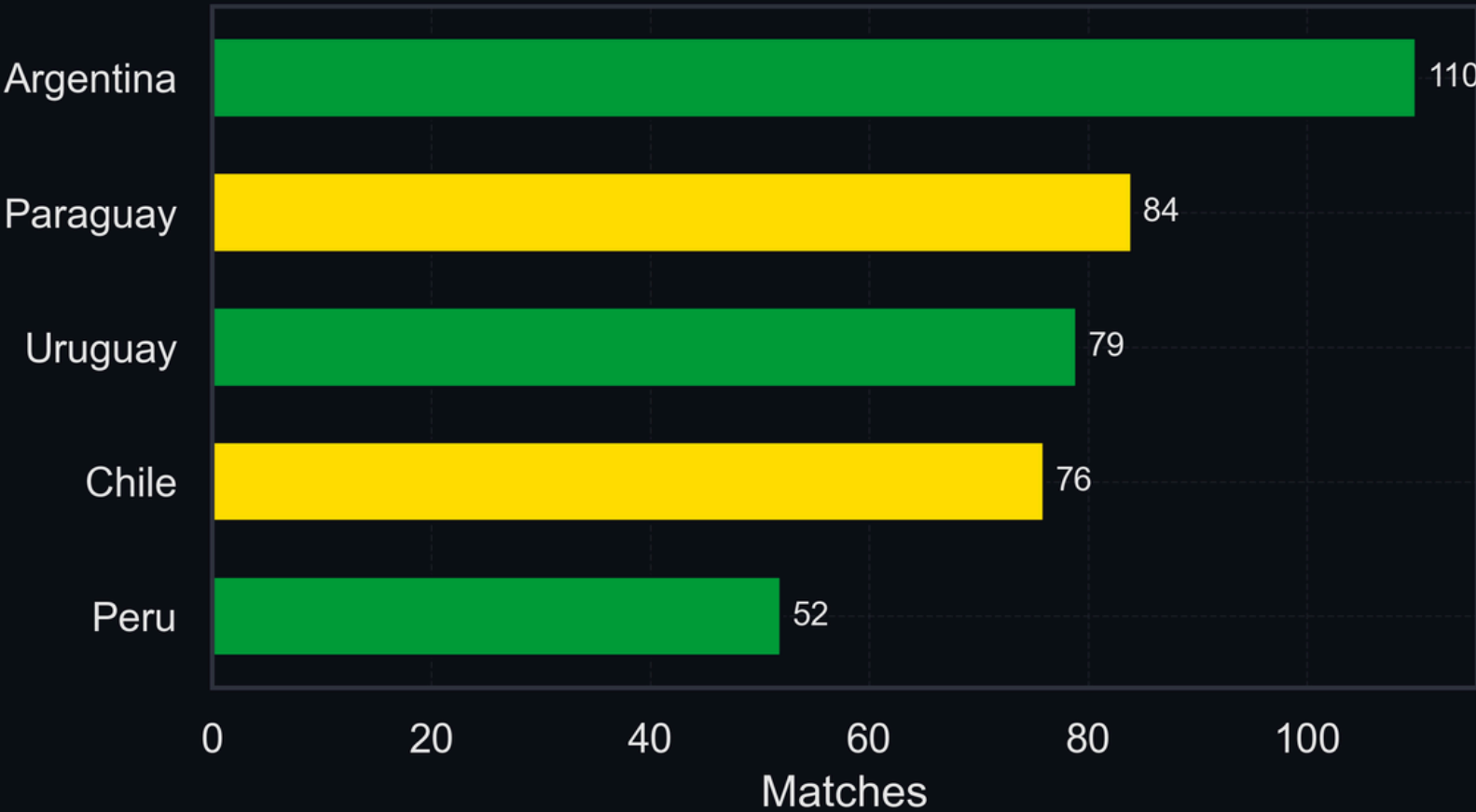
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

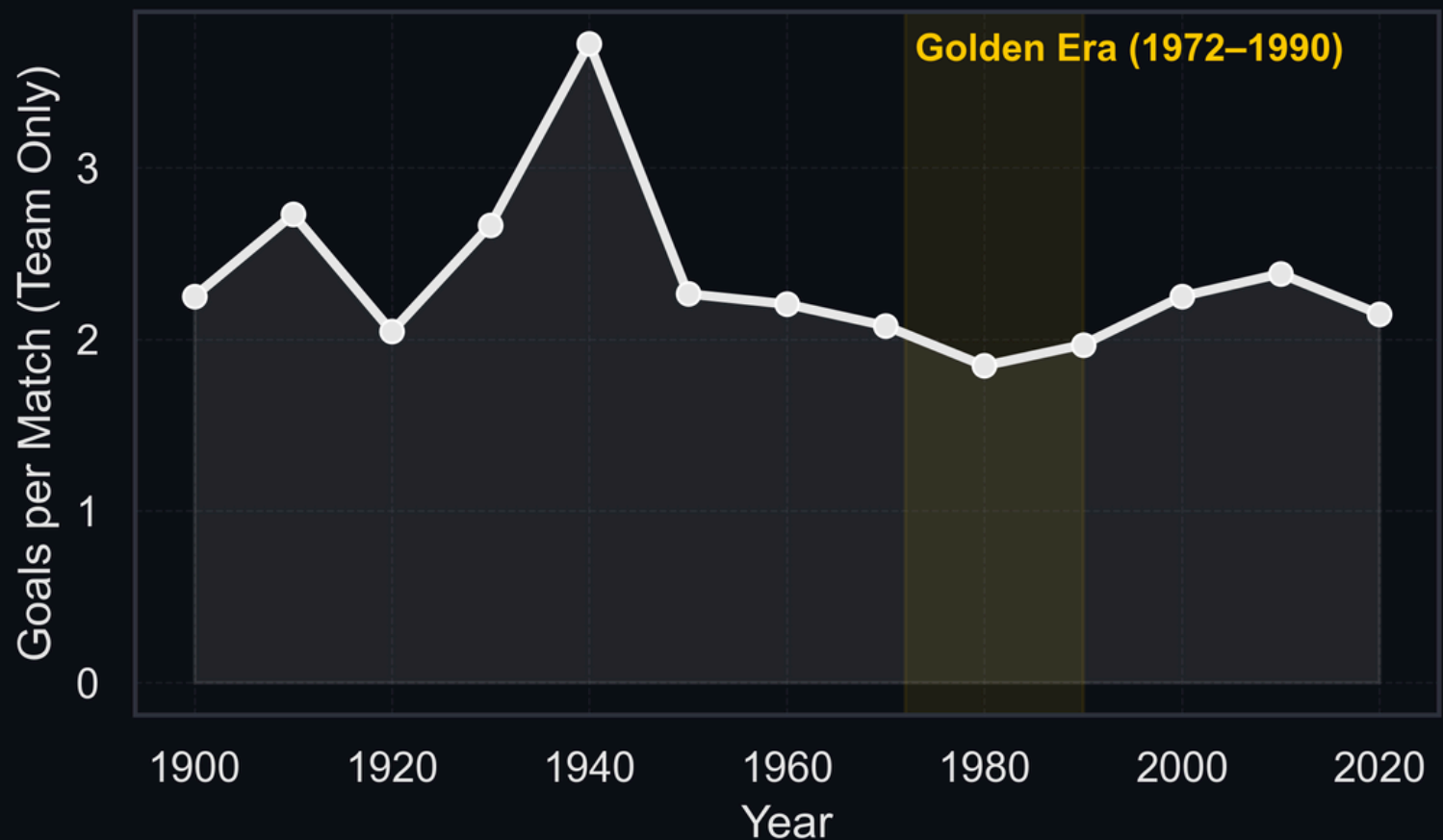


- 5× FIFA World Cup Titles
- ~63.5% Win Rate (1872–2020)
- Golden Era: 1958–2002
- Known for creativity and attacking flair

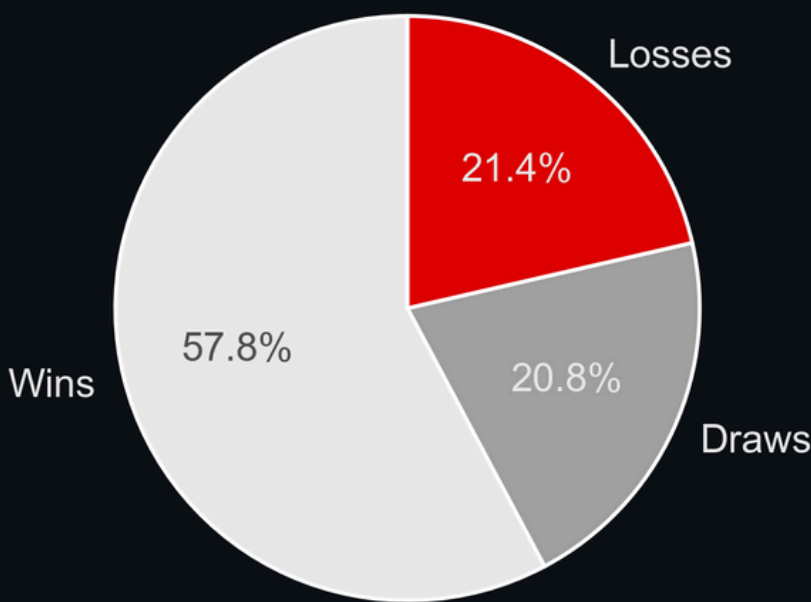
Average goals per match show Brazil's attacking evolution — reflecting the Seleção's consistent offensive dominance.

GERMANY

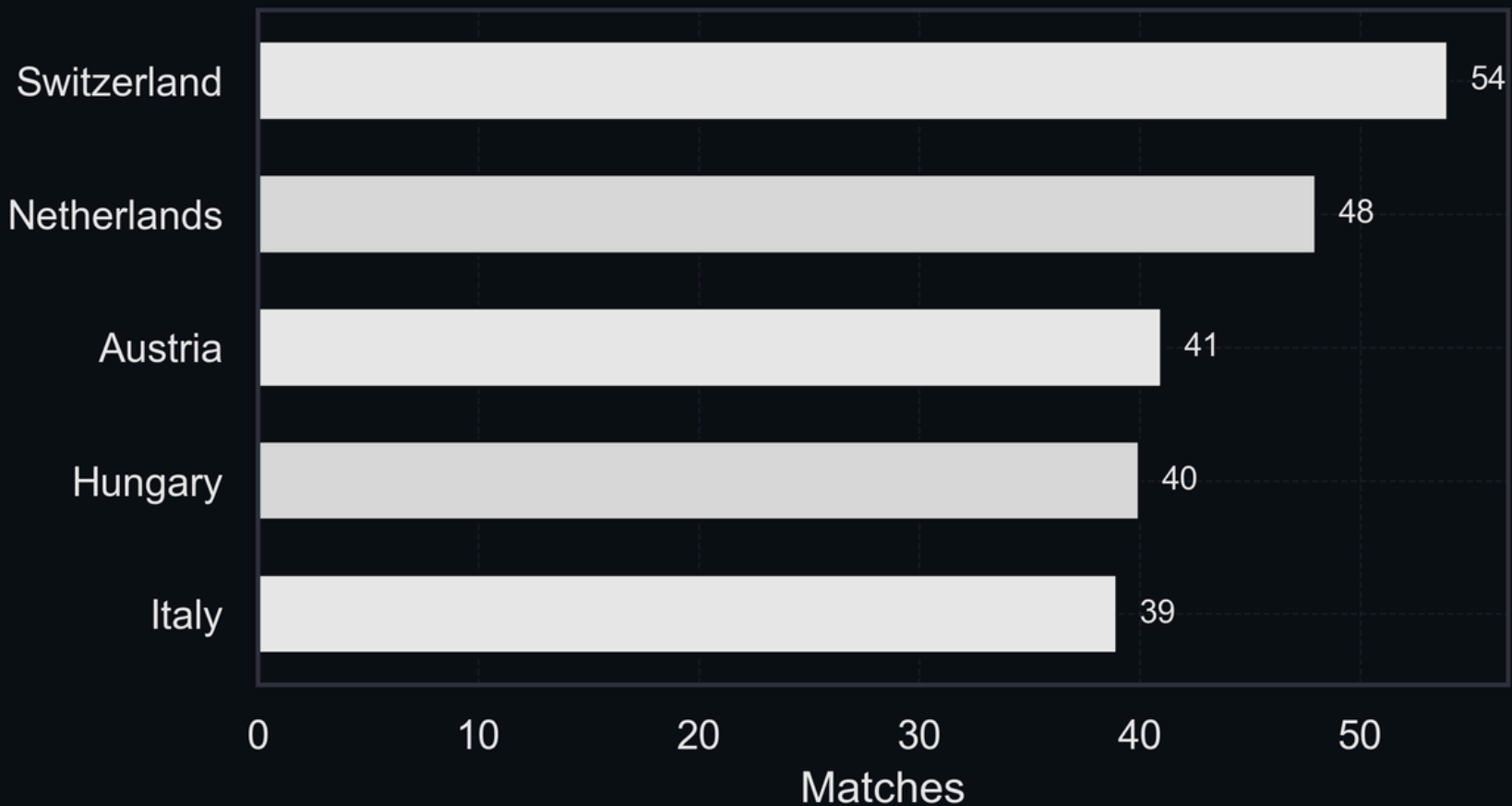
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

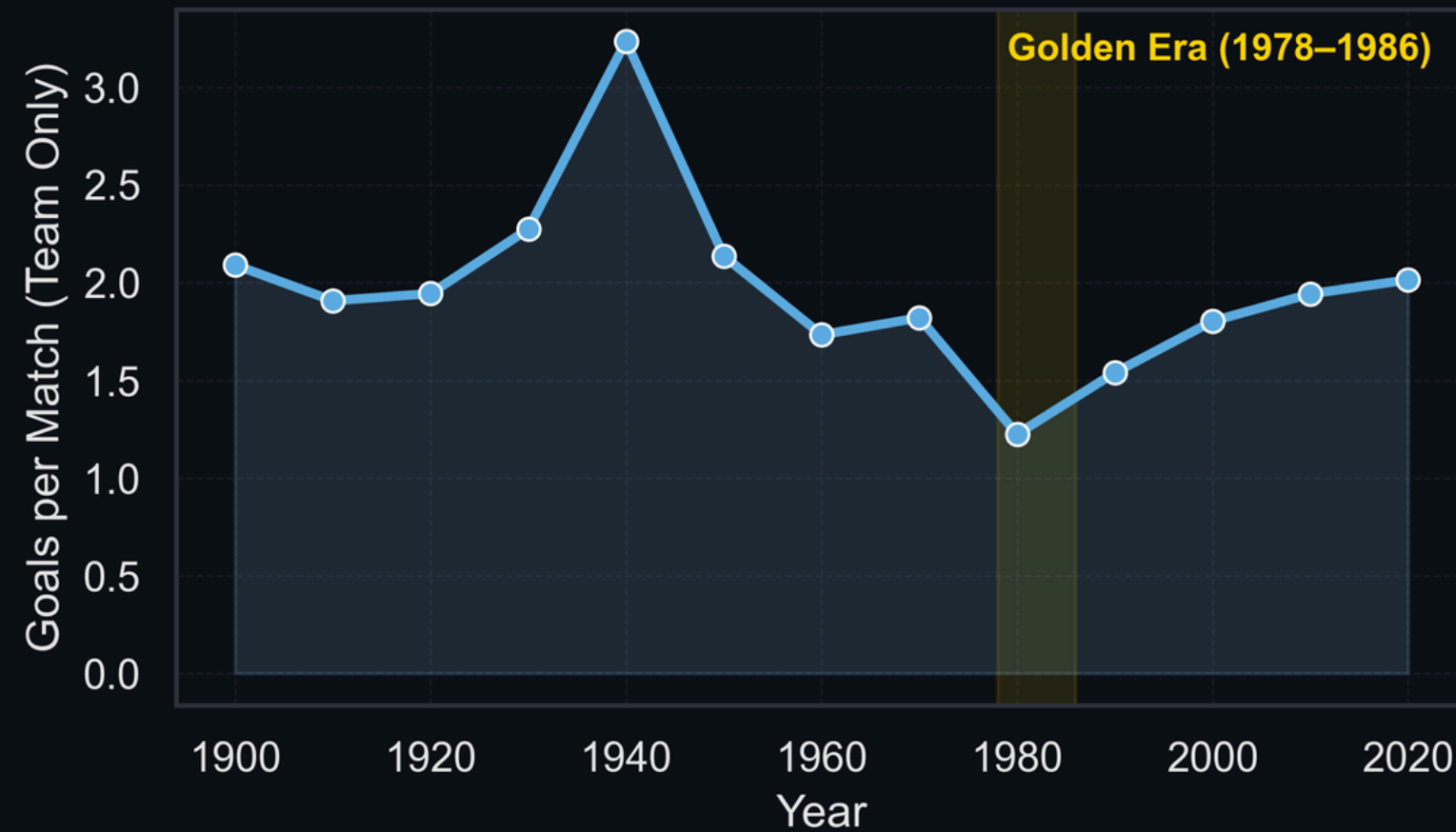


- 4× FIFA World Cup Titles
- 3× UEFA European Championships
- Golden Era: 1972–1990 (Beckenbauer–Matthäus Era)
- Known for efficiency, organization, and tactical strength

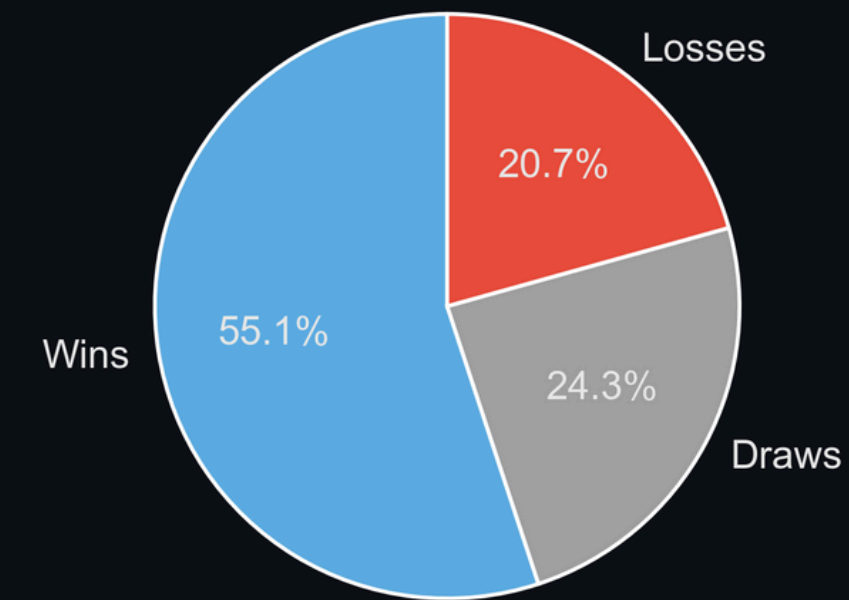
During this era, West Germany defined modern European football — combining structure, discipline, and collective mentality.

ARGENTINA

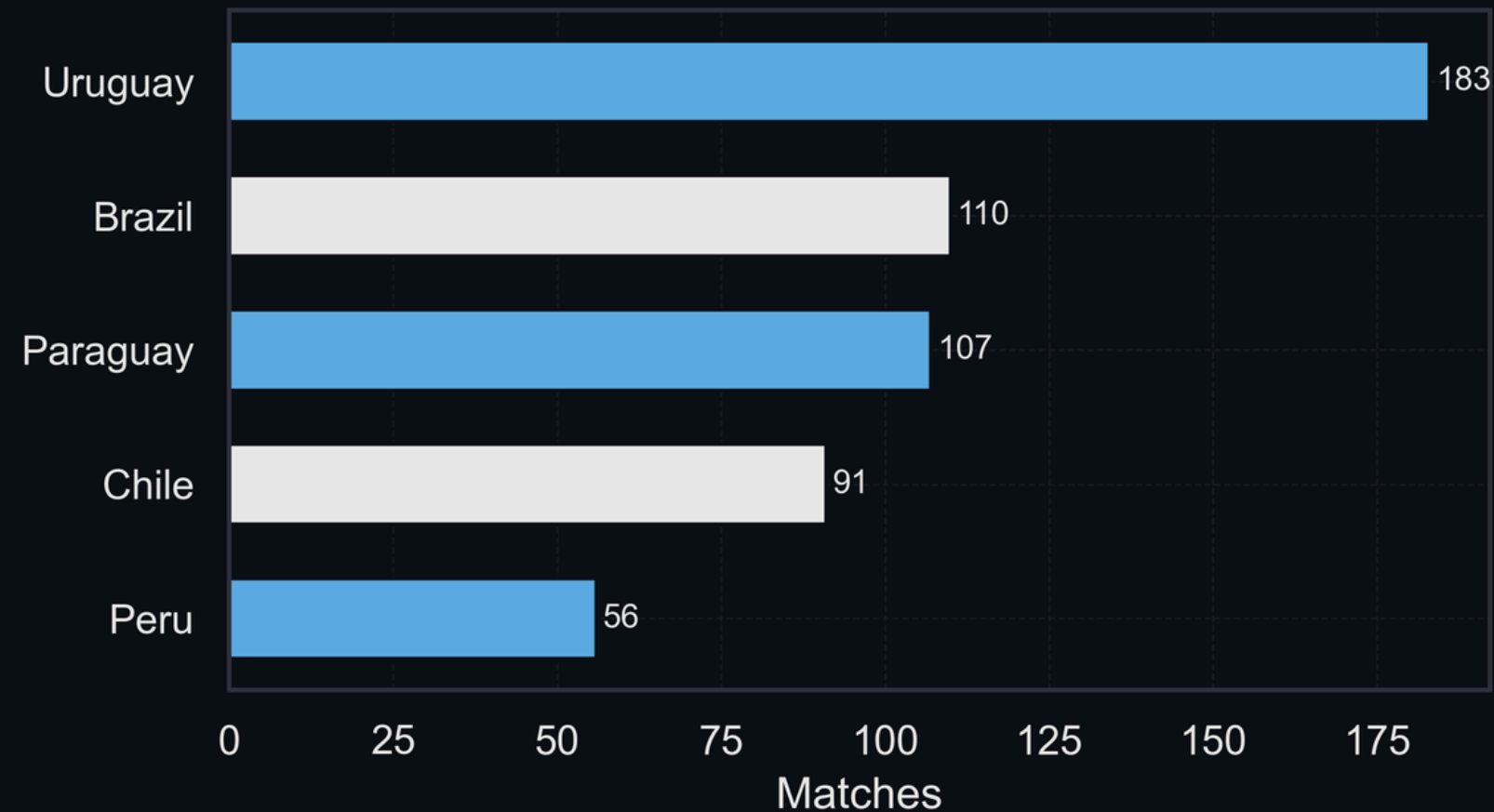
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

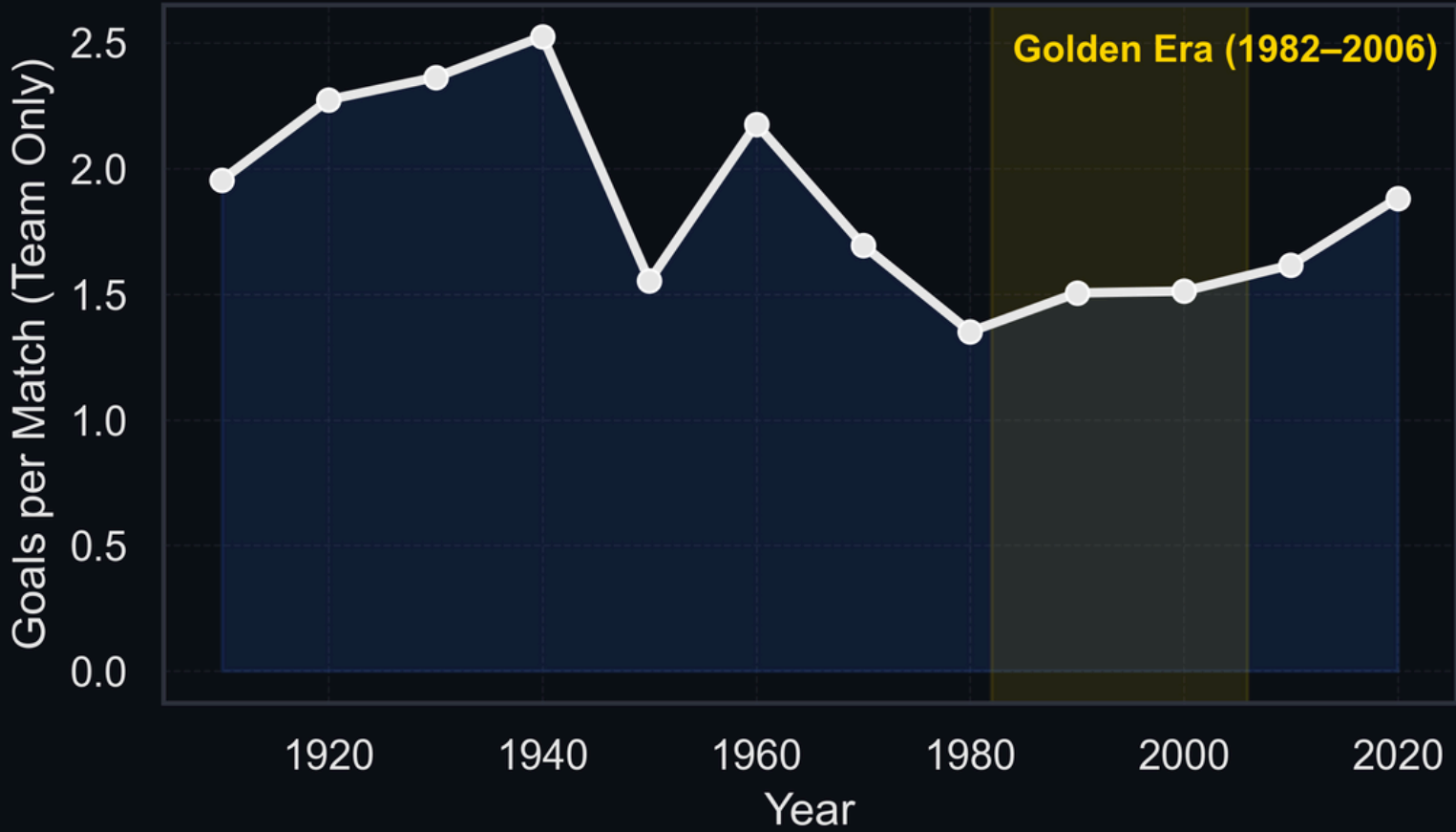


- 3× FIFA World Cup Titles (1978, 1986, 2022)
- 15× Copa América Titles
- Golden Era: 1978–1986 (Menotti–Maradona Era)
- Known for artistry, intensity, and passion

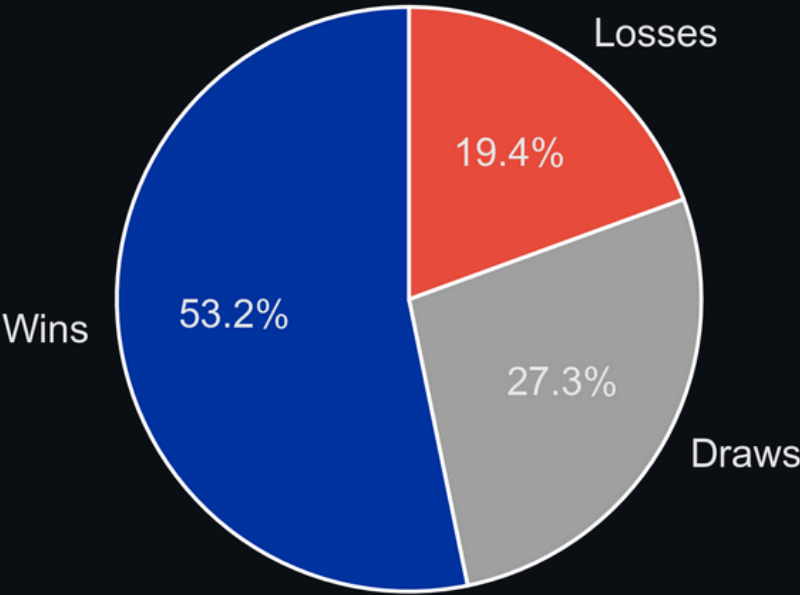
Between 1978 and 1986, Argentina reached global dominance — led by Maradona's brilliance and an unmatched emotional spirit.

ITALY

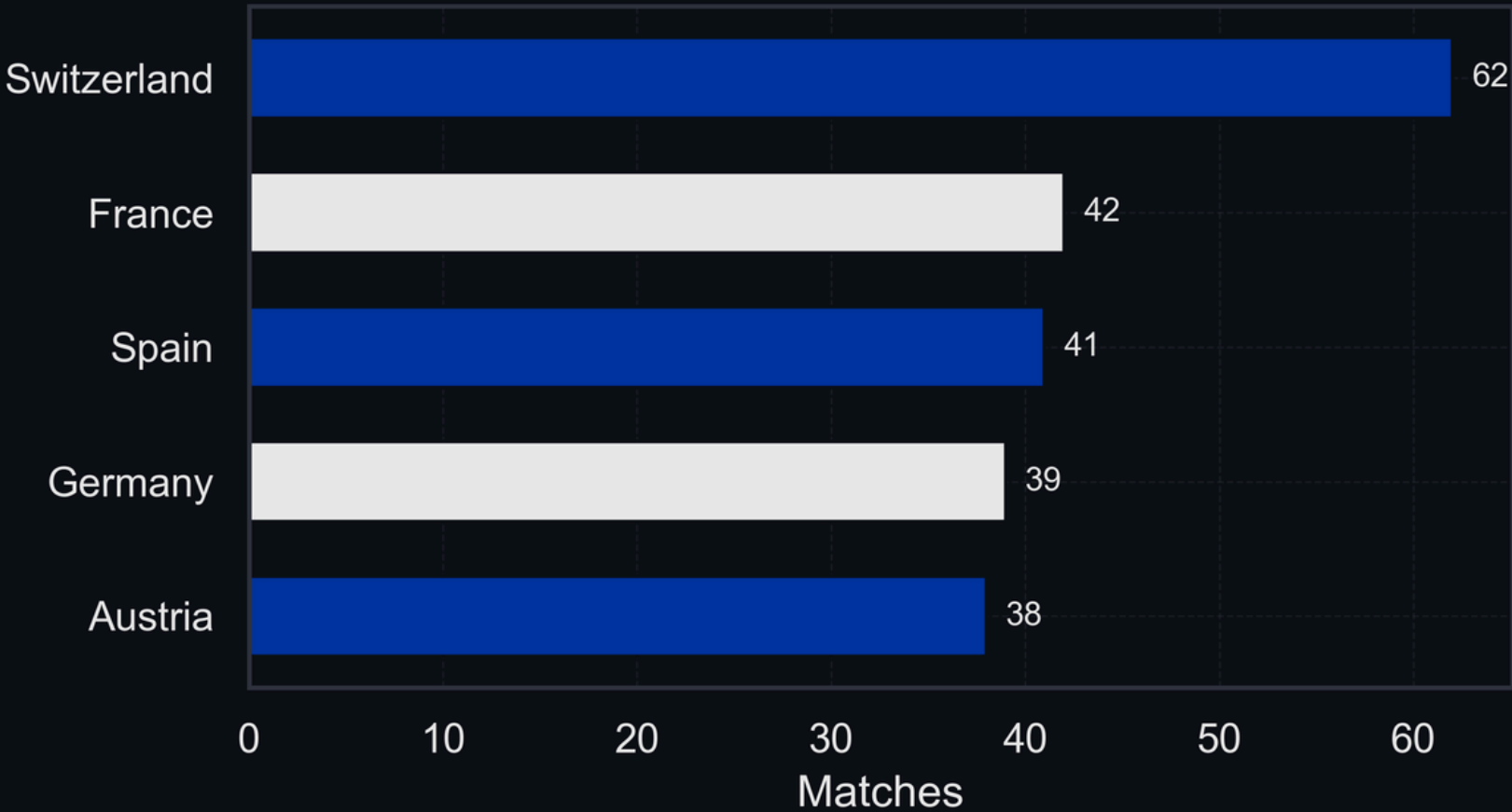
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

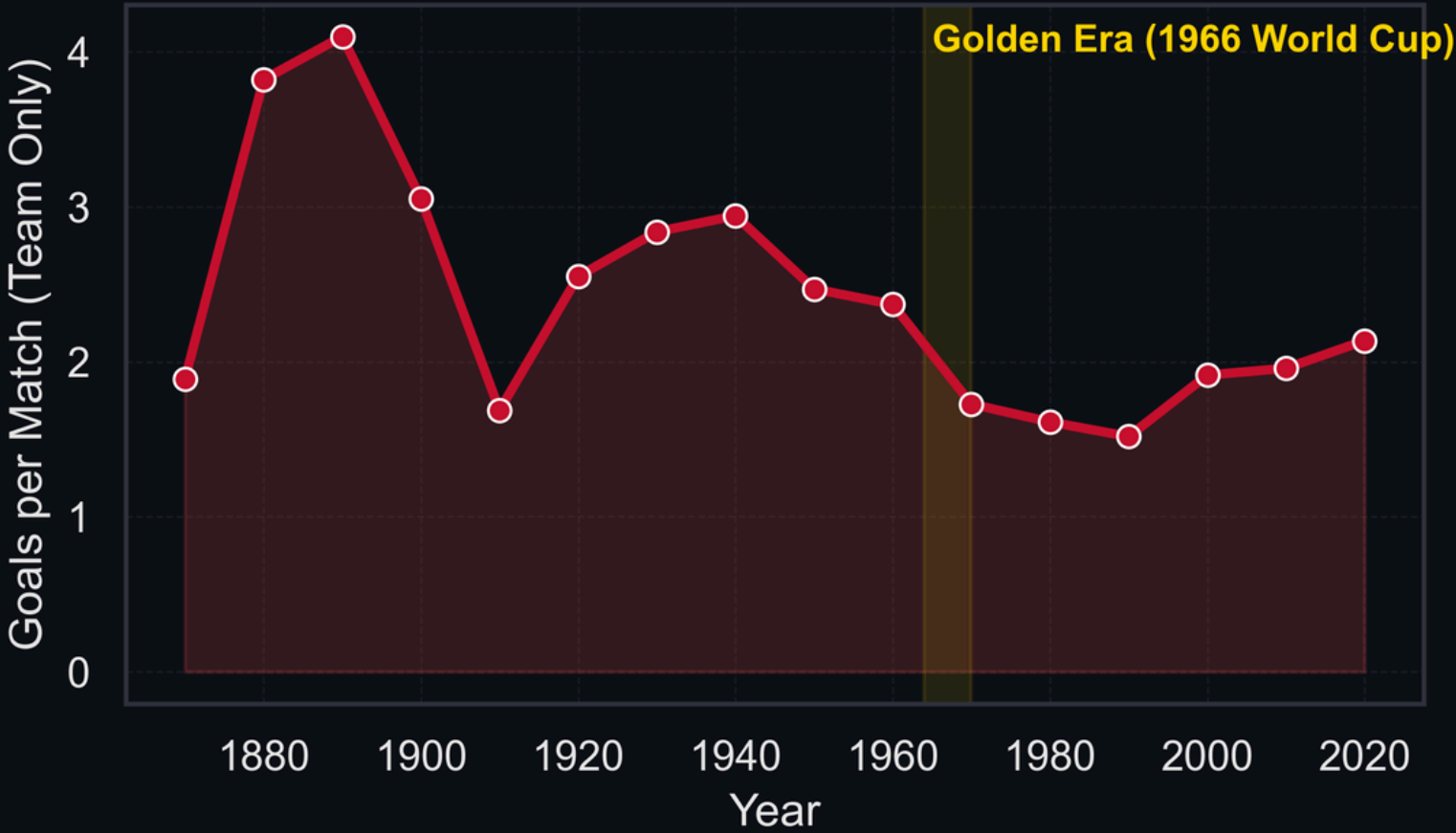


- 4× FIFA World Cup Titles (1934, 1938, 1982, 2006)
- 2× UEFA European Championship (1968, 2021)
- Golden Era: 1982–2006 (modern zirve)
- Known for tactical discipline, defensive mastery, and clutch mentality

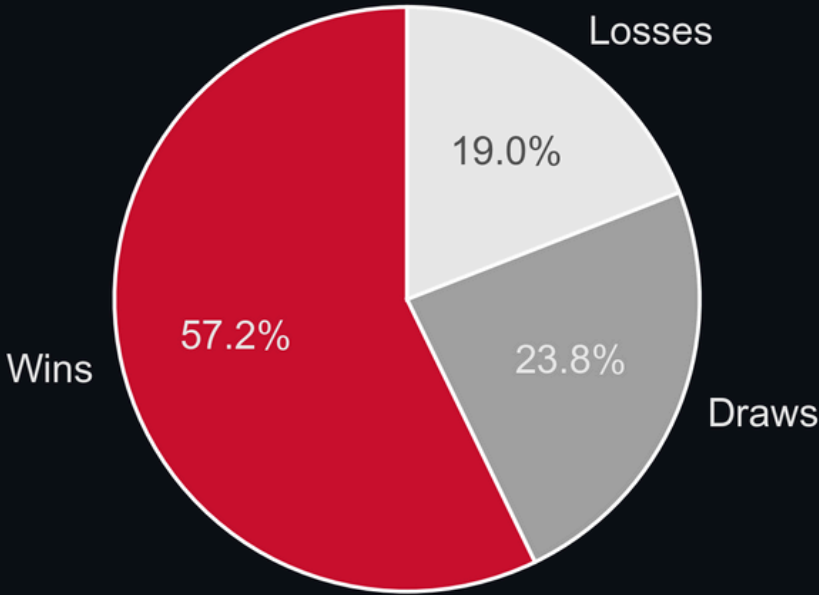
From 1982 to 2006, Italy combined elite defending with tournament nous — emerging as a modern-era powerhouse.

ENGLAND

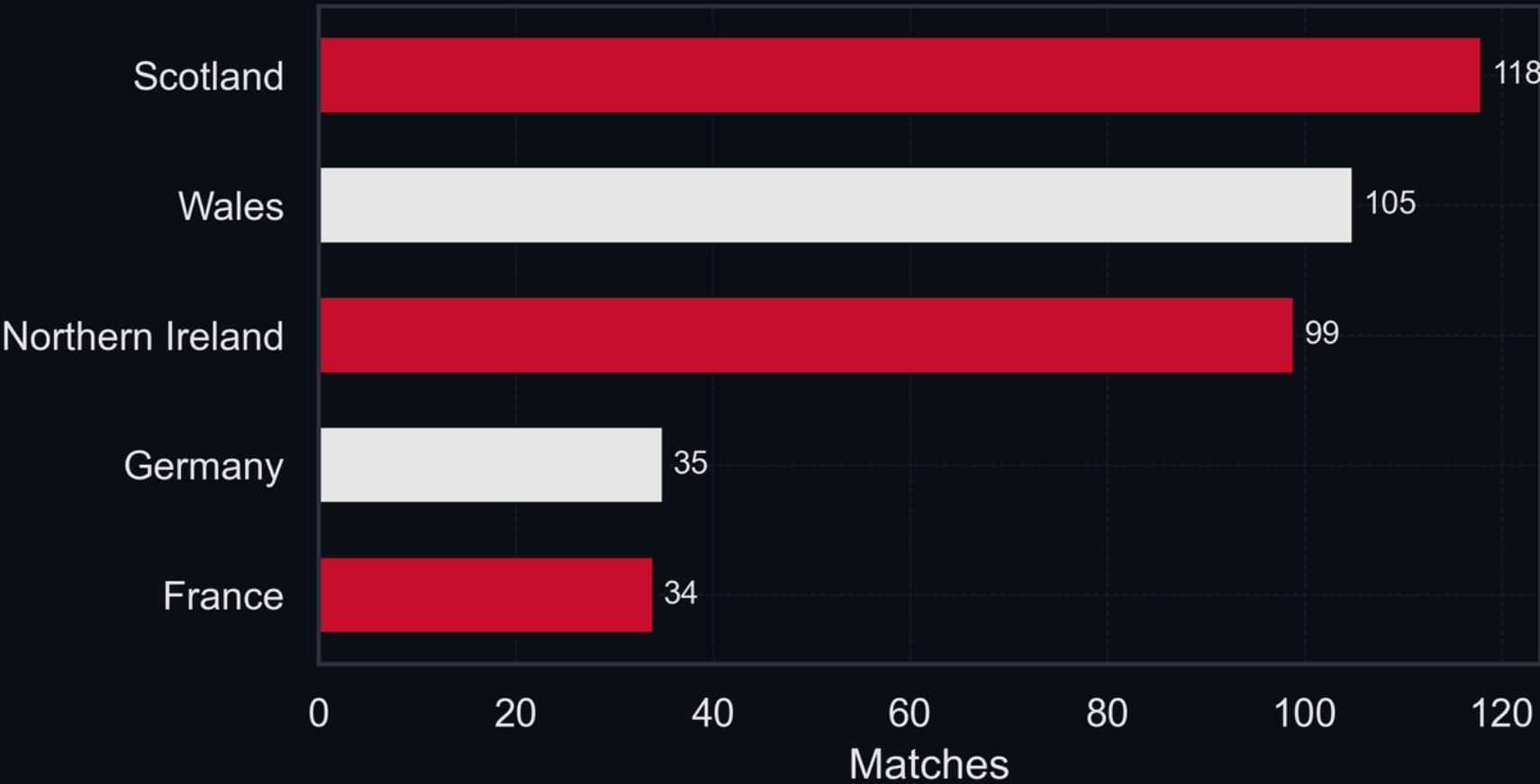
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

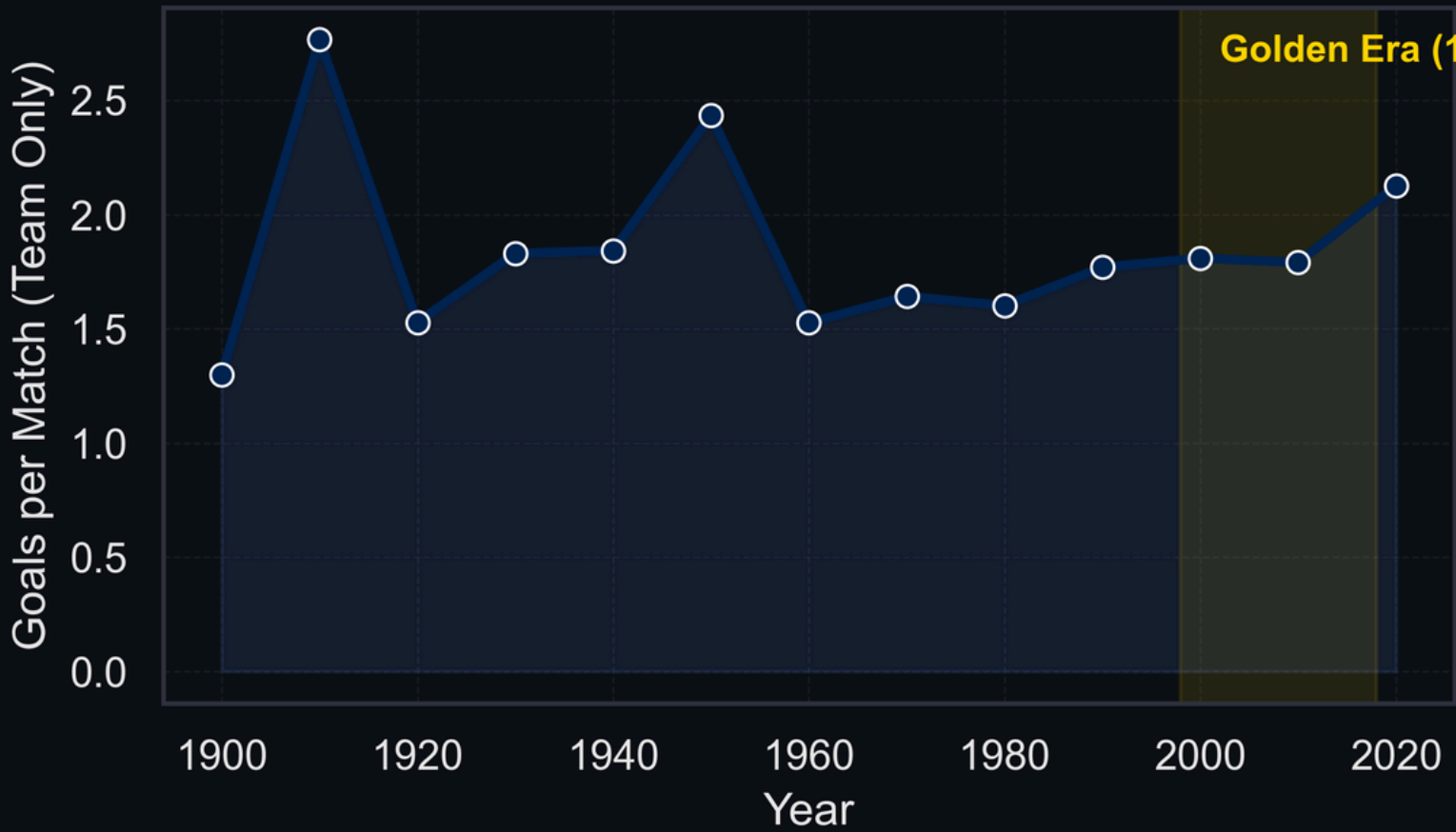


- 1× FIFA World Cup Title (1966)
- 2× UEFA Euro Runners-up (2020, 2024)
- Golden Era: 1966 (Charlton–Moore–Hurst generation)
- Known for passion, endurance, and aerial dominance

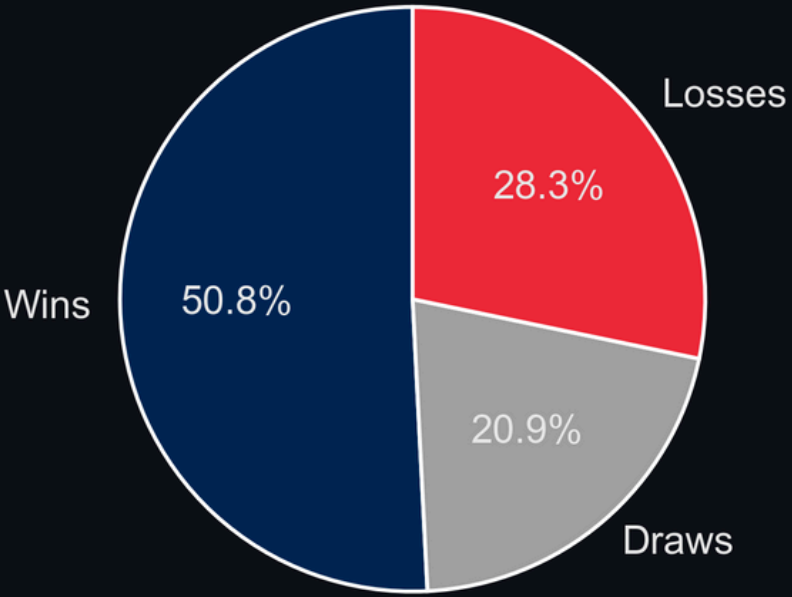
England, the birthplace of football, built its legacy on heritage, discipline, and iconic rivalries — defining the global game's spirit.

FRANCE

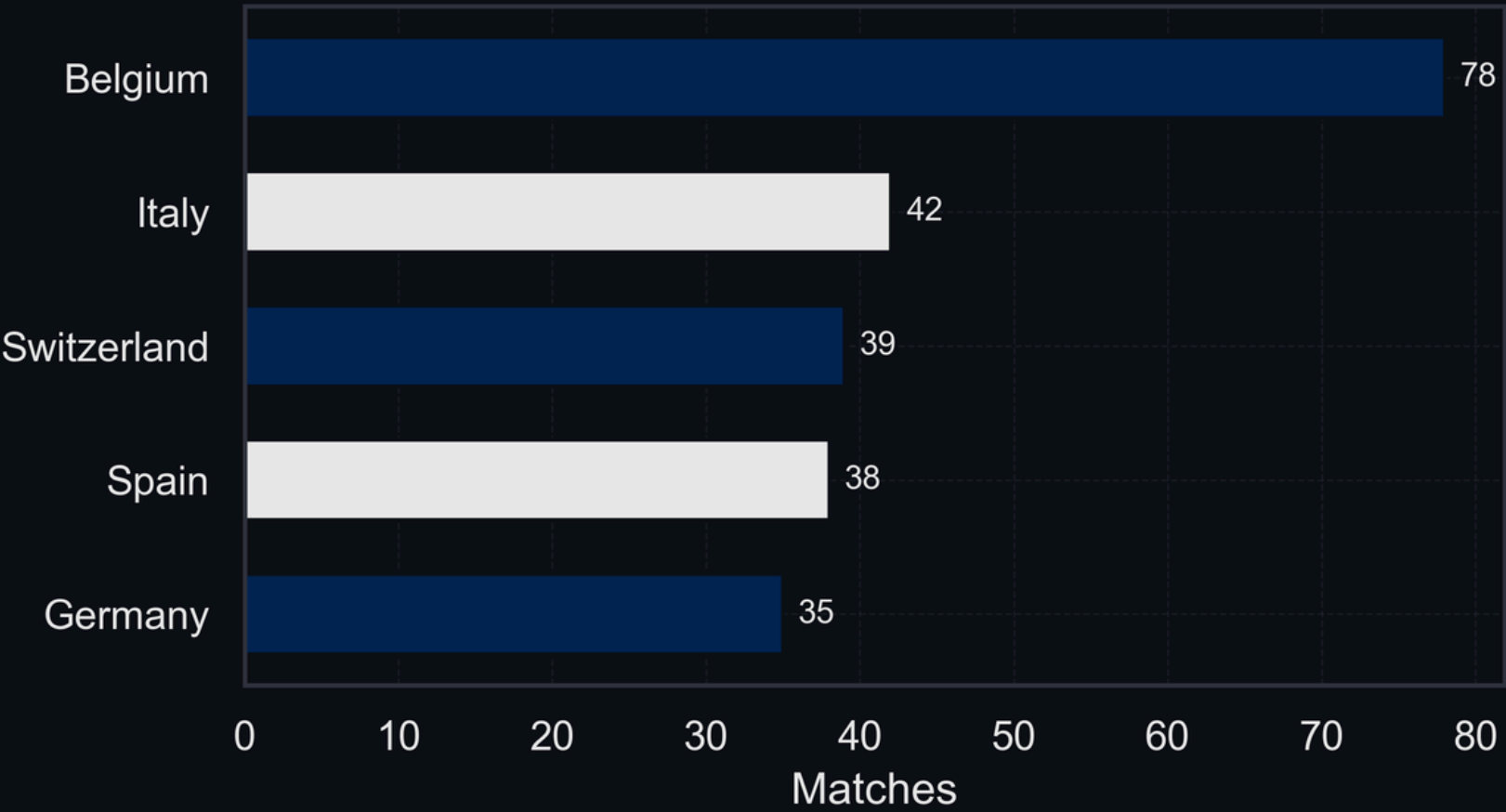
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

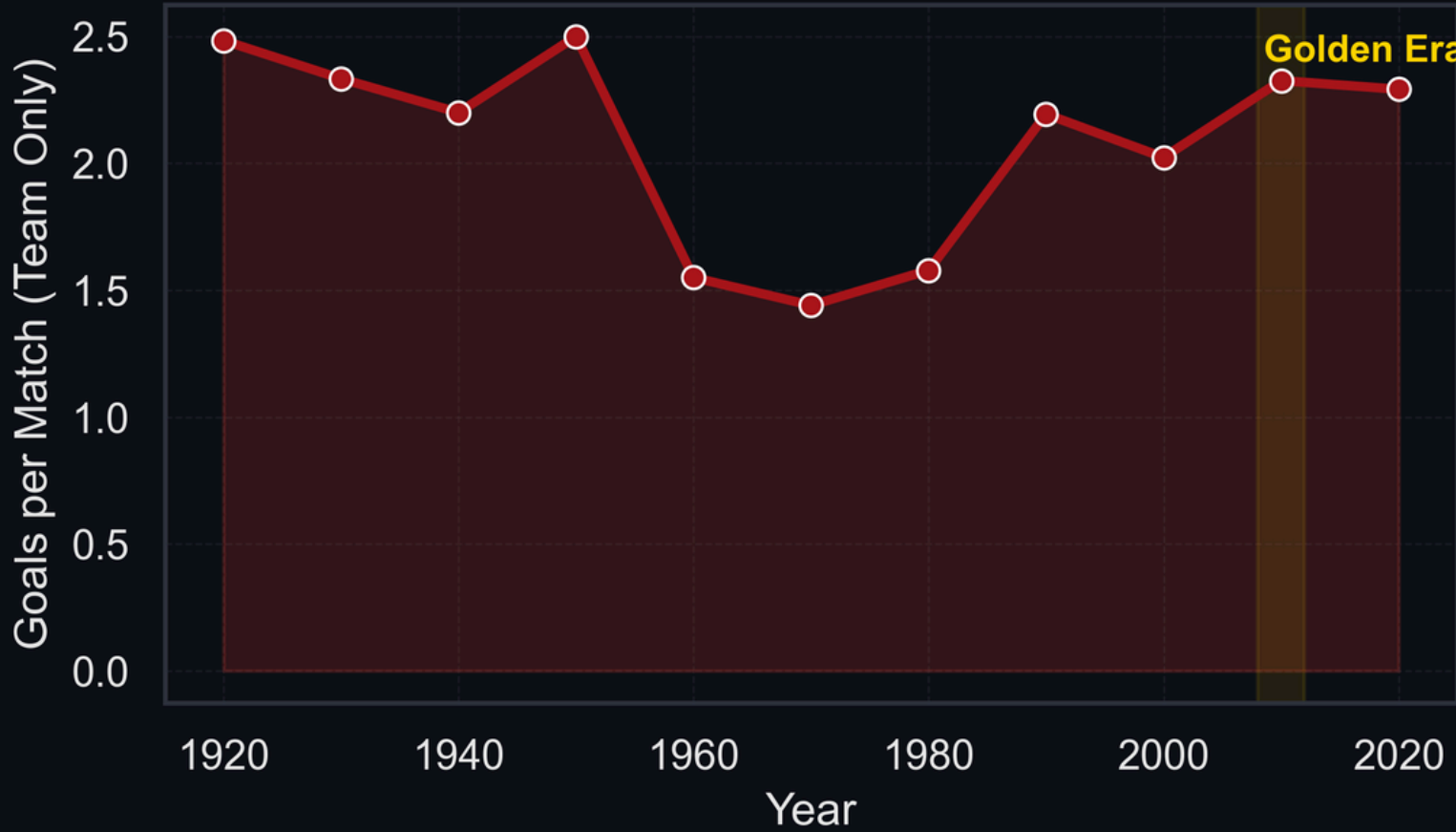


- 2× FIFA World Cup Titles (1998, 2018)
- 2× UEFA European Championship (1984, 2000)
- Golden Era: 1998–2018
- Known for balance, athleticism, and tactical flexibility

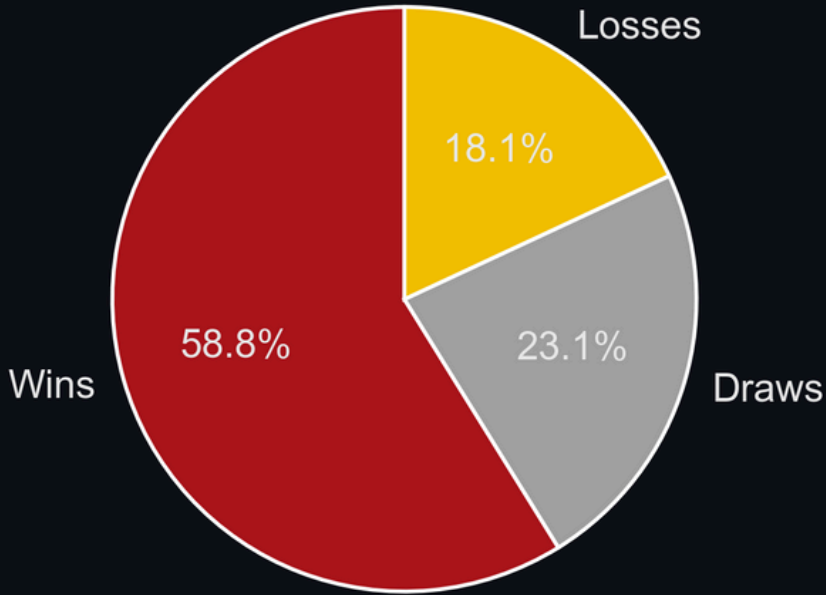
Between 1998 and 2018, France transformed into a modern football dynasty — combining individual brilliance with collective strength.

SPAIN

Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

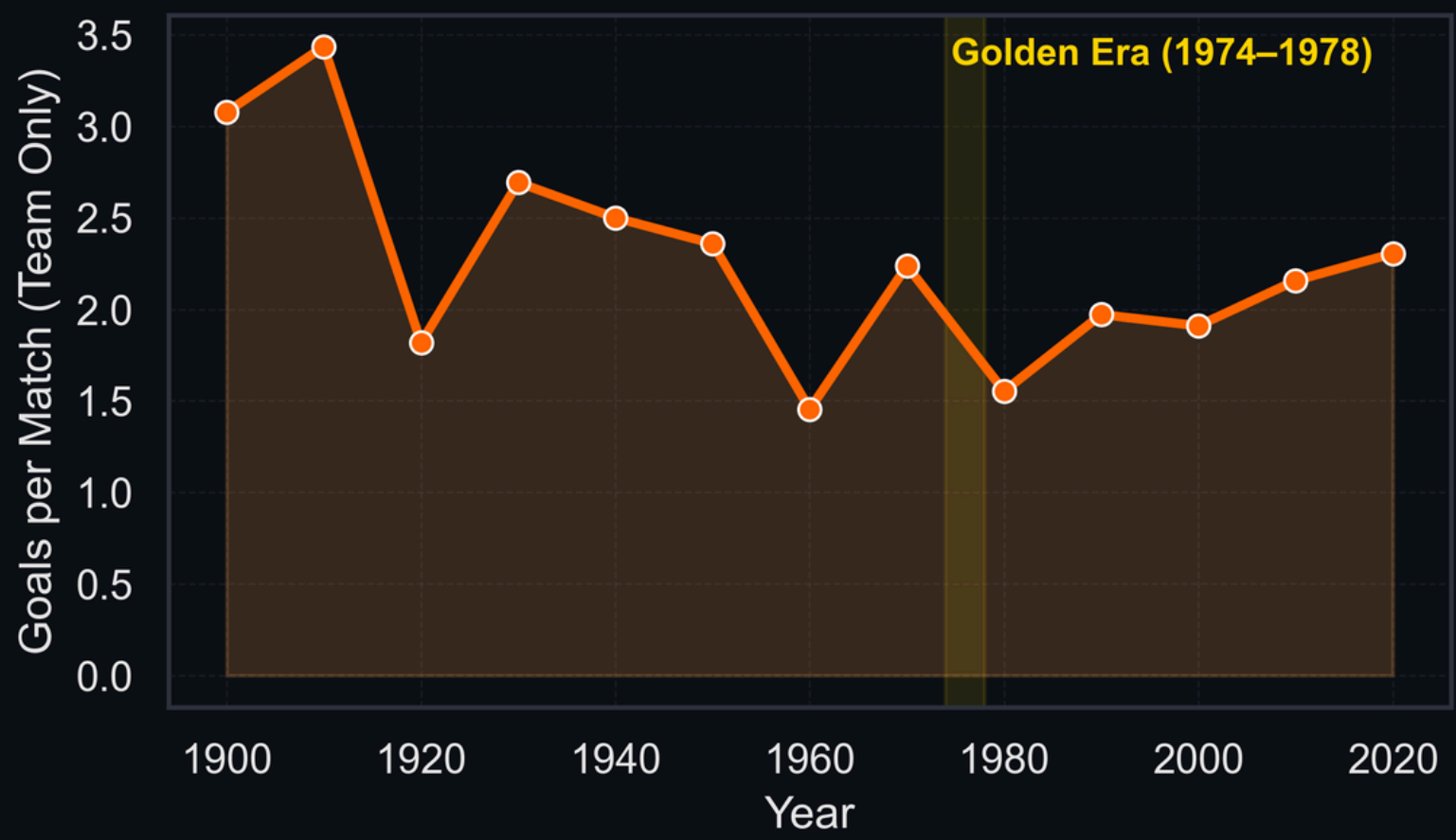


- 1× FIFA World Cup Title (2010)
- 3× UEFA European Championships (1964, 2008, 2012)
- Golden Era: 2008–2012 (Tiki-Taka era)
- Known for possession, precision, and technical mastery

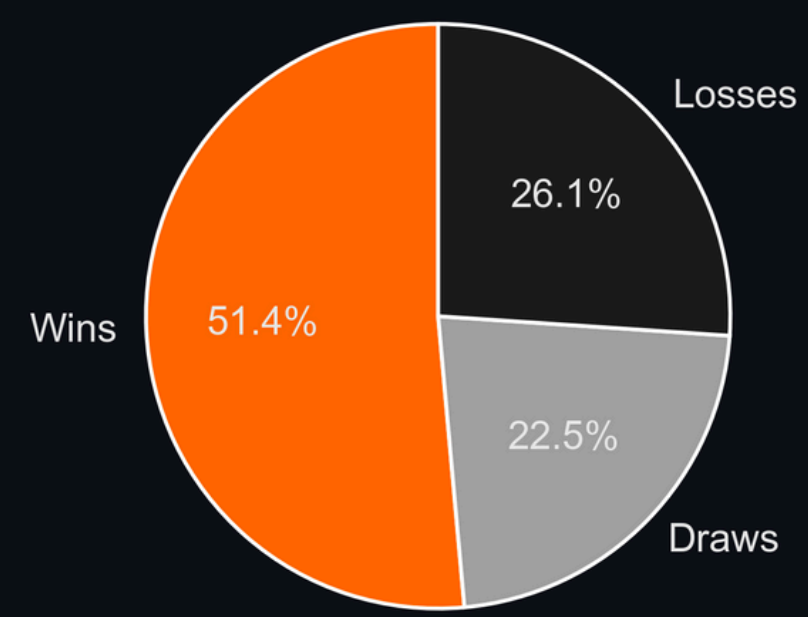
Between 2008 and 2012, Spain dominated world football — winning three consecutive major tournaments through unparalleled passing and control.

NETHERLANDS

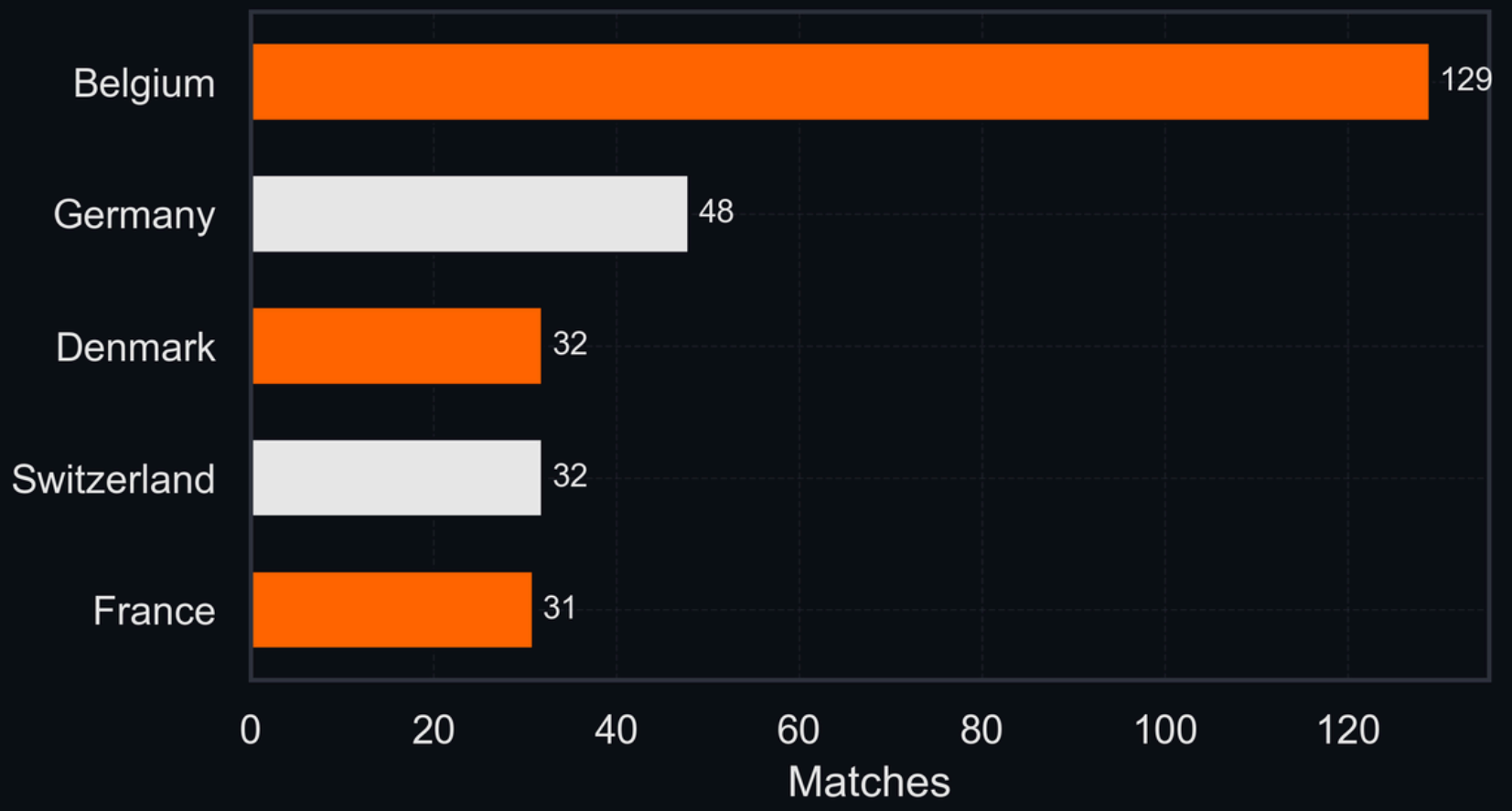
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches



- 3× FIFA World Cup Runners-up (1974, 1978, 2010)
- 1× UEFA European Championship (1988)
- Golden Era: 1974–1978 (Total Football)
- Known for fluid movement, tactical intelligence, and style

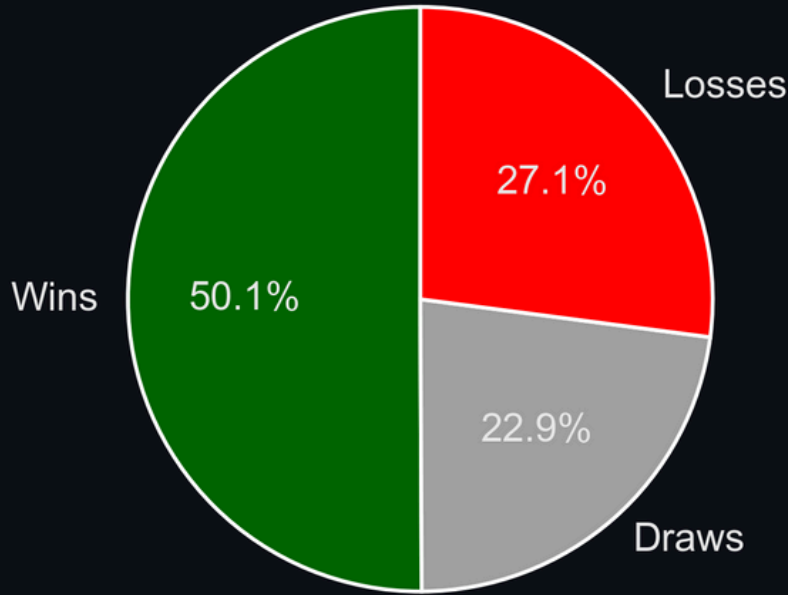
The Netherlands revolutionized football under Rinus Michels and Johan Cruyff — introducing 'Total Football', a philosophy that reshaped the global game.

PORTUGAL

Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches

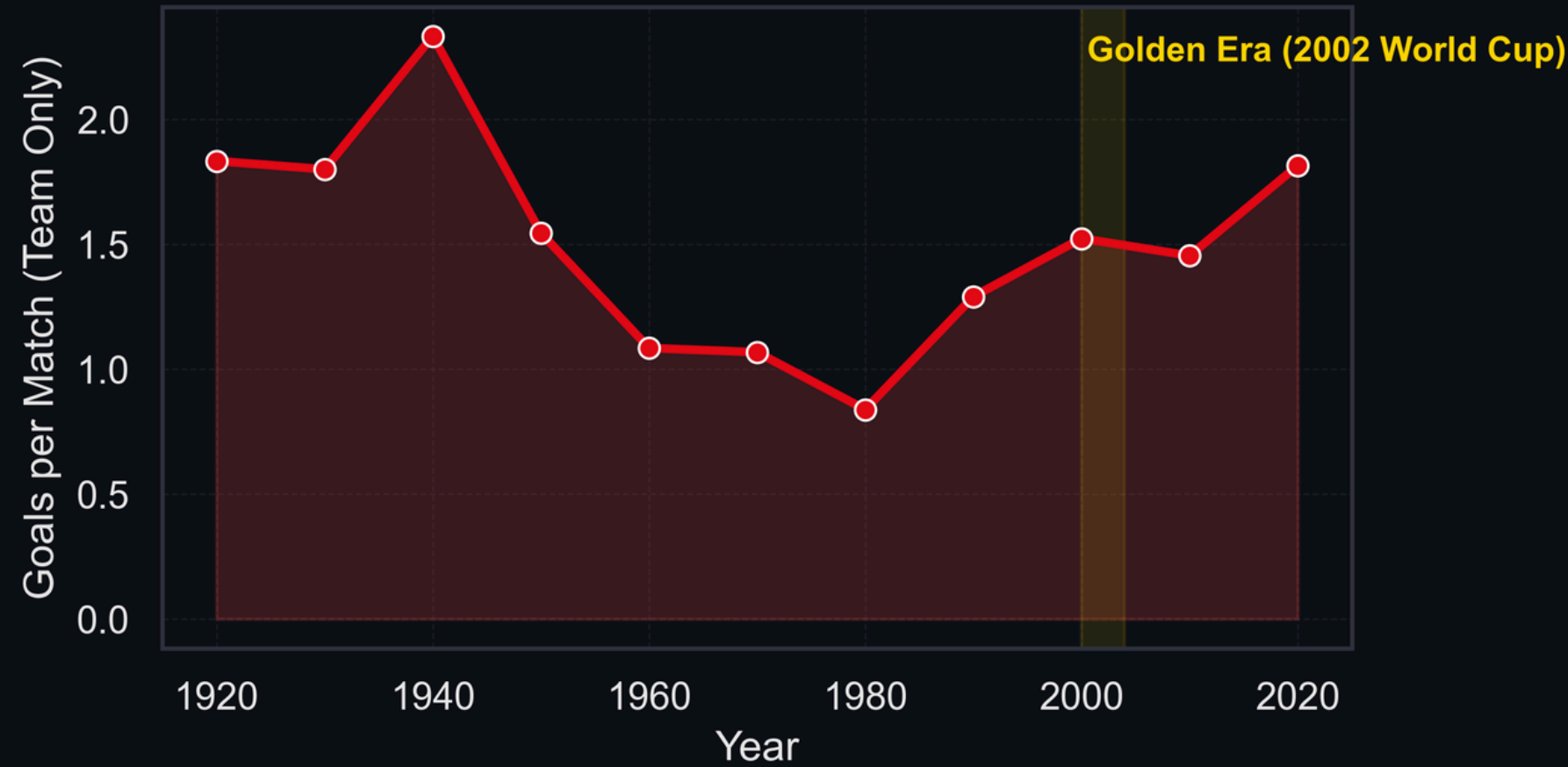


- 1× UEFA European Championship (2016)
- 1× UEFA Nations League (2019)
- Golden Era: 2004–2016 (Cristiano Ronaldo Era)
- Known for resilience, skill, and counter-attacking power

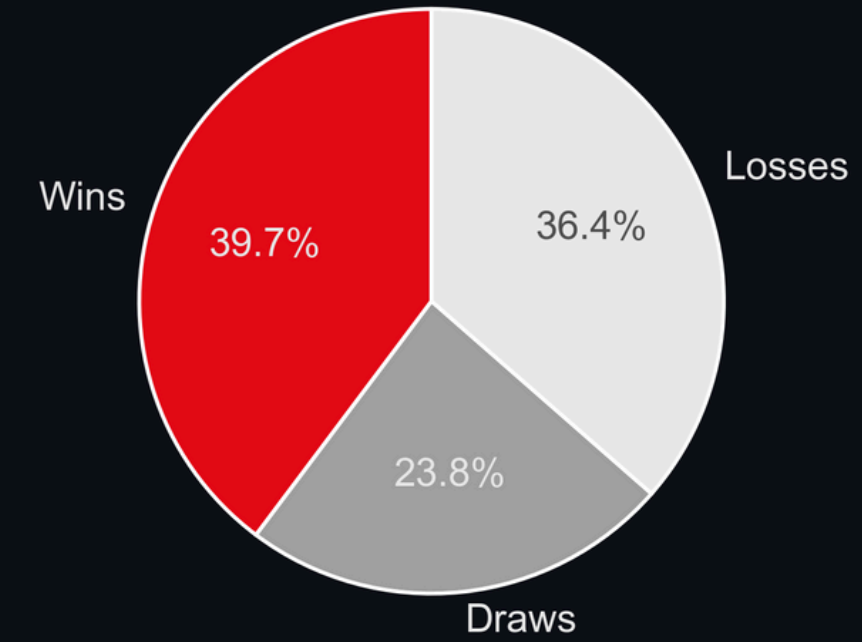
From 2004 to 2016, Portugal evolved into a modern European force — combining technical flair with defensive organization under Cristiano Ronaldo's leadership.

TÜRKİYE

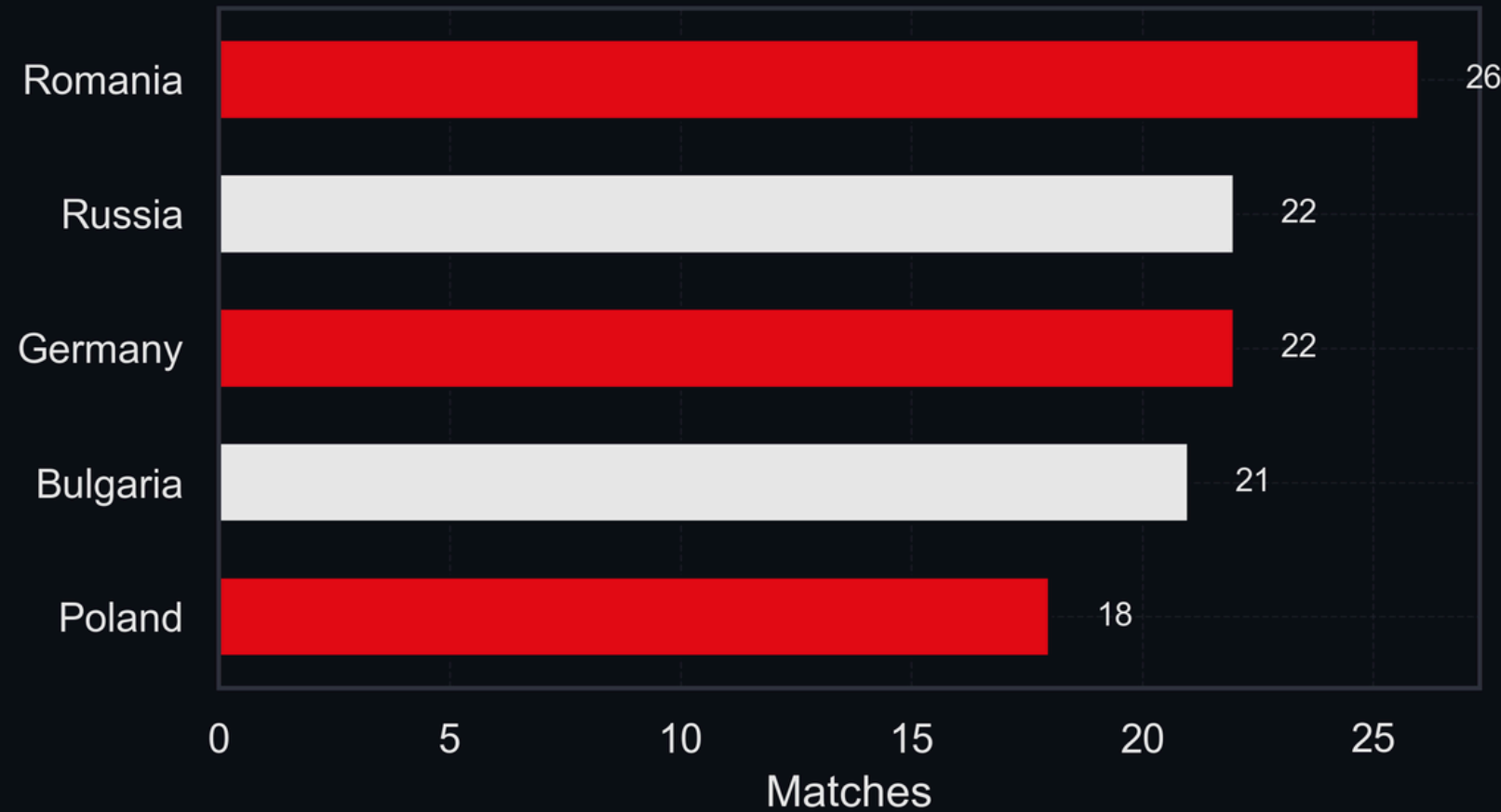
Goals per Match (Decadal Trend)



Overall Match Outcomes



Top 5 Opponents by Matches



- 3× FIFA World Cup appearances (1954, 2002, 2022)
- 5× UEFA European Championship appearances
- Golden Era: 2002 (World Cup 3rd Place)
- Known for passion, unpredictability, and fighting spirit

In 2002, Türkiye shocked the world by finishing 3rd at the World Cup — showcasing determination, unity, and national pride on the global stage.

This project visualized over 150 years of international football data to uncover global and national trends in scoring, competitiveness, and tournament evolution.

The results highlight football's transformation from a regional pastime into a global phenomenon — shaped by tactical evolution, cultural identity, and enduring rivalries.